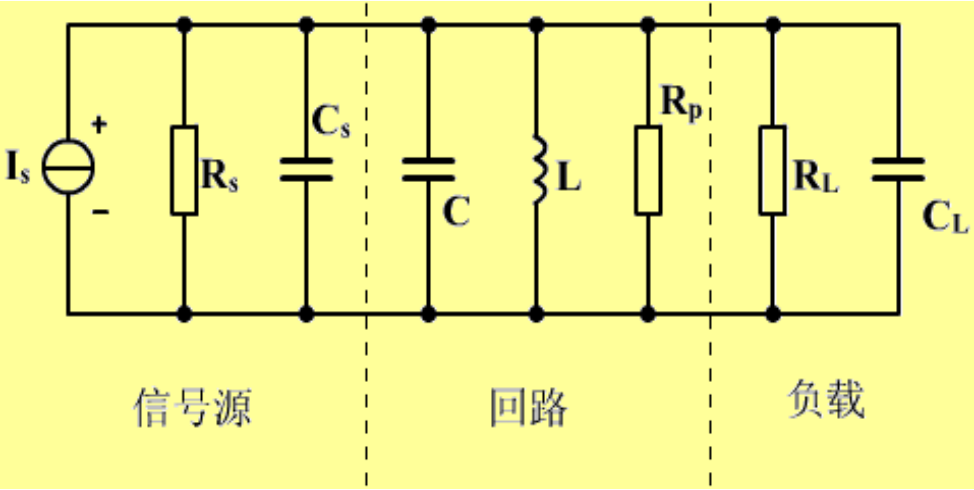
Integrated Communication Circuits

复习习题课

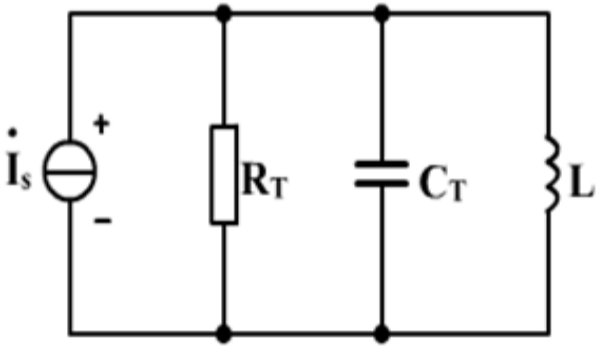
阻抗变换

信号源内阻及负载对谐振回路的影响。



图(a)

等效 →



图(b)

（b）图中：$R_T = R_S // R_P // R_L \downarrow Q_L = R_T / \omega_0 L \downarrow . < Q_0$

$$C_T = C_S + C + C_L \uparrow \omega_0' = \frac{1}{\sqrt{LC_T}} \downarrow . < \omega_0$$

谐振回路的前一级（或信号源）

下一级（或负载）的阻抗

对谐振回路的性能影响较大

增益下降

选择性变差

谐振频率漂移

解决这一问题的方法之一：

采用部分接入法，

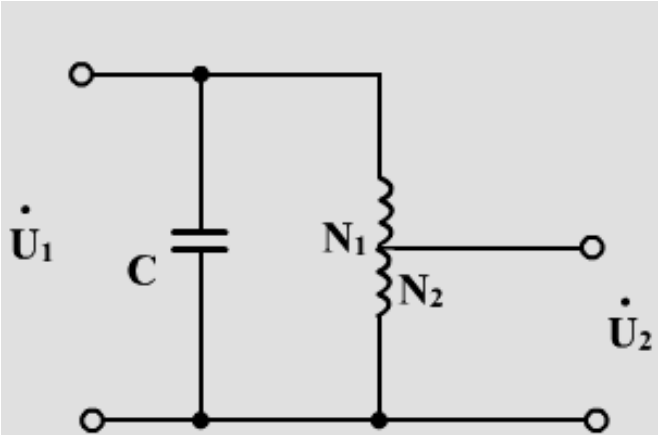
减小谐振回路的前一级（或信号源）和下一级（或负载）的阻抗对谐振回路的影响。

部分接入的电路：

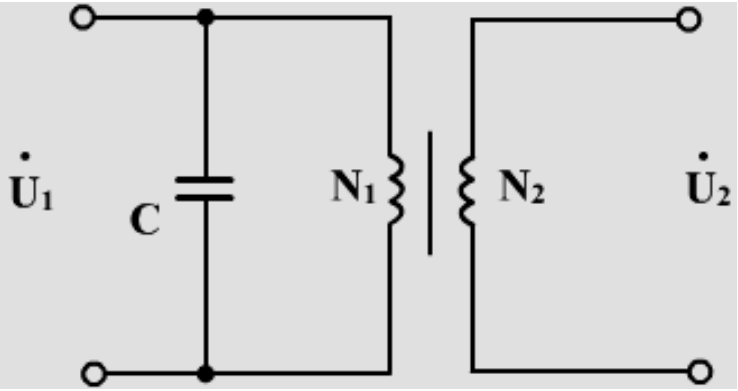
- 变压器耦合
- 电感分压
- 电容分压

(1)接入系数 p

定义为接入电压U_2与回路总电压U_1的比值

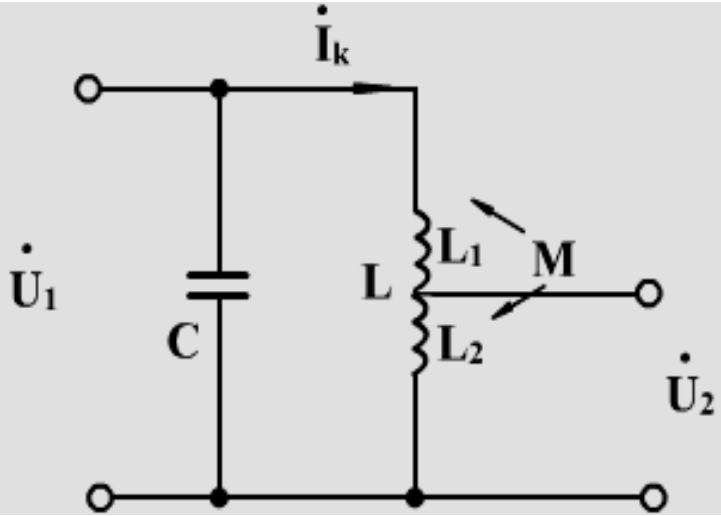


图(a)自耦变压器.



图(b)变压器.

图(a)(b)的接入系数：$p = \dfrac{N_2}{N_1}$



图(c)：

$$p=\frac{\dot{U}_2}{\dot{U}_1}=\frac{\dot{I}_k j\omega(L_2\pm M)}{\dot{I}_k j\omega(L_1+L_2\pm M)}$$

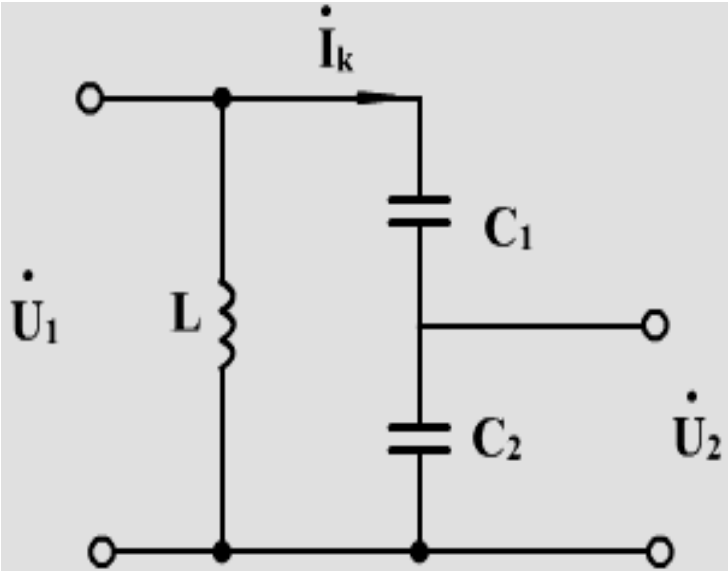
$$=\frac{L_2\pm M}{L}$$

图(c)电感分压

$$\text{总电感}: L=L_1+L_2\pm 2M.$$

$$\text{若}M=0 \qquad p=\frac{L_2}{L}=\frac{L_2}{L_1+L_2}$$

图(d)



图(d)电容分压

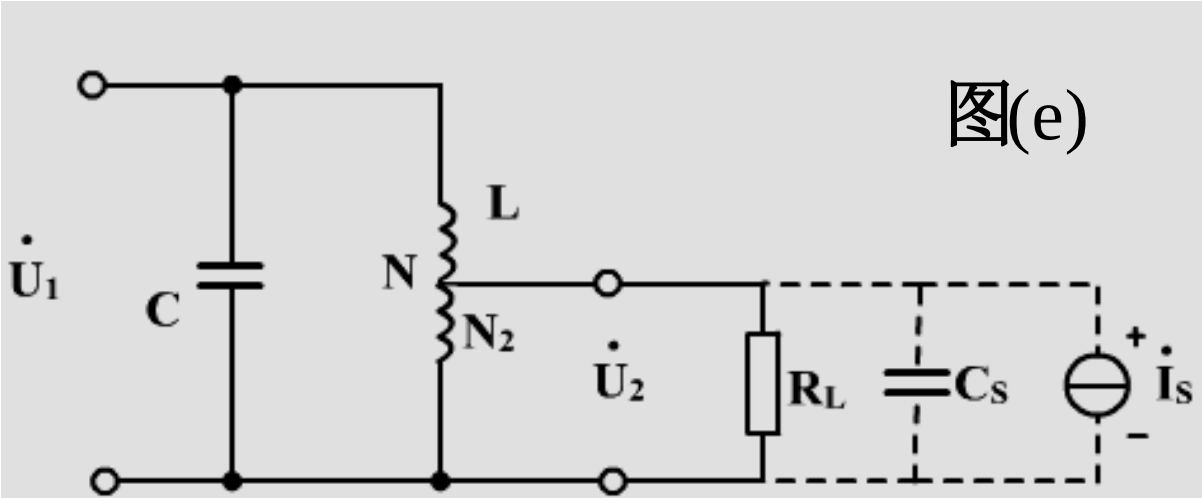
$$p=\frac{\dot{U}_2}{\dot{U}_1}=\frac{\dot{I}_k\dfrac{1}{j\omega C_2}}{\dot{I}_k\dfrac{1}{j\omega C}}$$

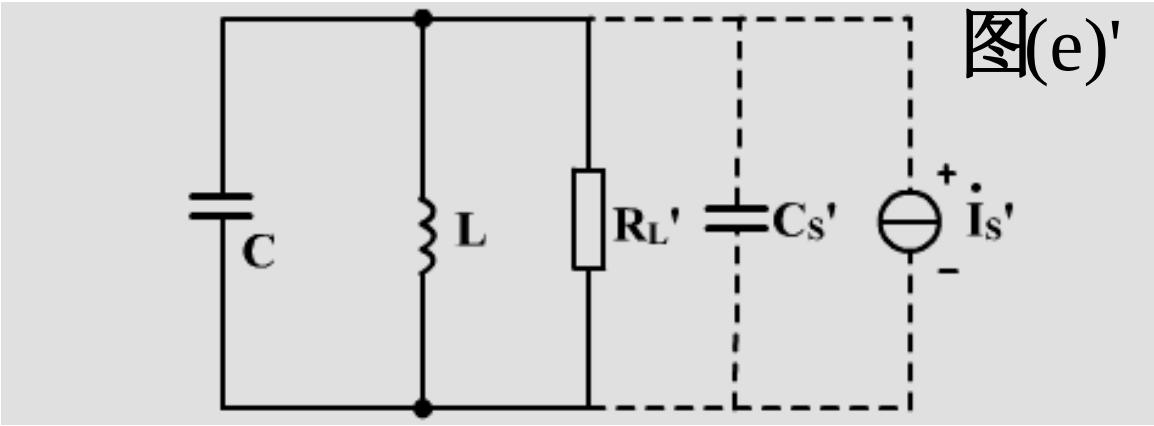
$$=\frac{C}{C_2}$$

$$\text{总电容}:C=\frac{C_1C_2}{C_1+C_2}$$

$$\therefore p=\frac{C_1}{C_1+C_2}$$

(2)负载阻抗的部分接入:





(e)图R_L吸收功率：$\dfrac{U_2^{\ 2}}{R_L}$

$(e)'$图R_L'吸收功率：$\dfrac{U_1^{\ 2}}{R_L'}$

两者相等：$\dfrac{U_2^{\ 2}}{R_L}=\dfrac{U_1^{\ 2}}{R_L'}$

$$\therefore R_L'=\left(\frac{U_1}{U_2}\right)^2 R_L=\frac{R_L}{p^2}$$

同理：$C_s'=p^2C_s,\qquad \dot{I}_s'=p\dot{I}_s$

例2

(1)在调谐放大器的负载回路两端并联电阻，试问对放大器的谐振增益、通 频带、选择性等各有什么影响?

(2)若调谐放大器的回路电感 L的损耗电阻增大，重复以上问题.

解

$$\text{并联谐振阻抗}R_T \downarrow \rightarrow A_{V0} \downarrow$$

$$Q_L = \omega_0 C R_T \downarrow \rightarrow BW \uparrow$$

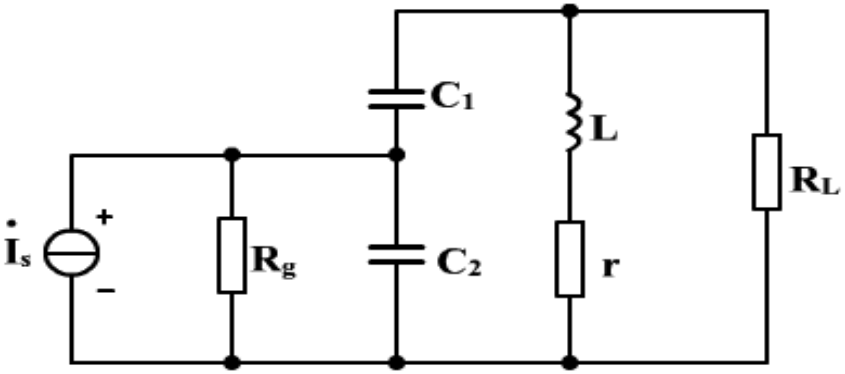
$$S = \frac{1}{\sqrt{1+\xi^2}} = \frac{1}{\sqrt{1+(2Q_L \frac{\Delta\omega}{\omega_0})^2}} \uparrow$$

$S\uparrow$ 选择性变差， 而 $K_{0.1}=\dfrac{BW_{0.1}\uparrow}{BW\uparrow}$ 不变

（2） $r\uparrow\rightarrow R_{\mathrm{p}}=\dfrac{X_{\mathrm{s}}^{2}}{r}\downarrow\rightarrow R_{\mathrm{T}}\downarrow$ 影响同上.

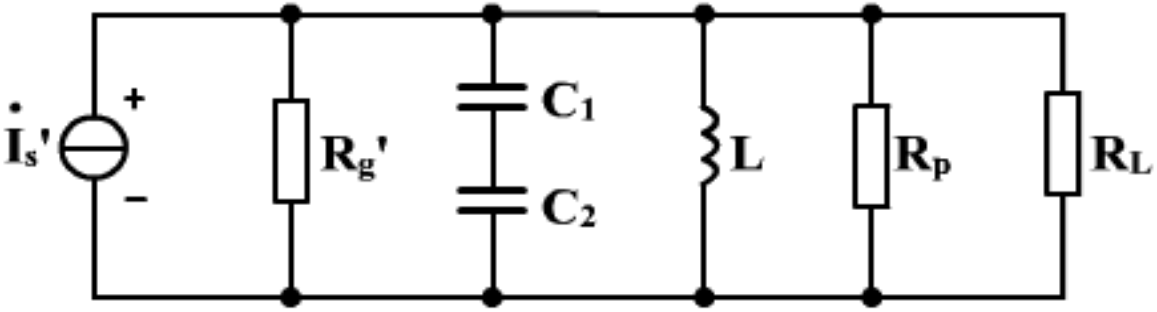
例3

题图的电容式分压并联谐振电路参数：L=200μH, r=8Ω, C_1=140pF, C_2=1400pF, R_g=5kΩ, R_L=100kΩ.
求 f_0 和 BW。



解

高频电路等效:



$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{L\dfrac{C_1C_2}{C_1+C_2}}} = \cdots = 1\text{MHz}$$

$$Q_0 = \frac{\omega_0 L}{r} = \frac{2\pi\times10^6\times200\times10^{-6}}{8} = 157$$

$$R_\text{p} = rQ_0{}^2 = \frac{L}{Cr} = 8\times157^2 = 197\text{k}\Omega$$

$$p = \frac{C_1}{C_1+C_2} = 0.091 \qquad R_\text{g}' = \frac{R_\text{g}}{P^2} = \frac{5\times10^3}{0.091^2} = 604\text{k}\Omega$$

$$R_{\Sigma} = R_{\mathrm{g}}' // R_{\mathrm{p}} // R_{\mathrm{L}} = 60\mathrm{k}\Omega$$

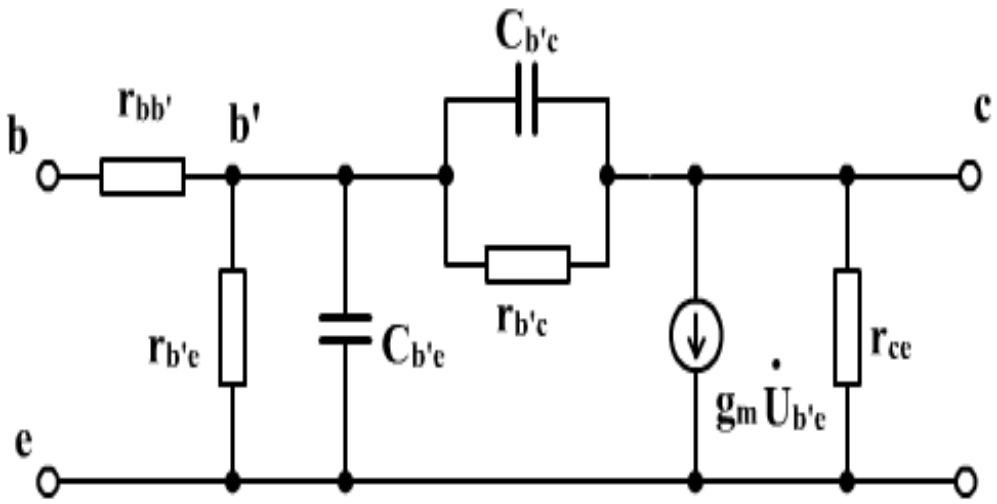
$$Q_{\mathrm{L}} = \frac{R_{\Sigma}}{\omega_0 L} = \frac{60\times10^3}{2\pi\times10^6\times200\times10^{-6}} = 47.8$$

$$\therefore BW = \frac{f_0}{Q_{\mathrm{L}}} = \frac{10^6}{47.8} = 20.92\mathrm{kHz}.$$

单级单回路调谐放大器

一、晶体管等效电路

1.混合π参数等效电路.

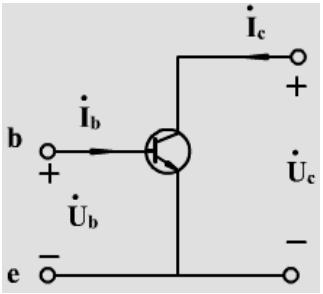


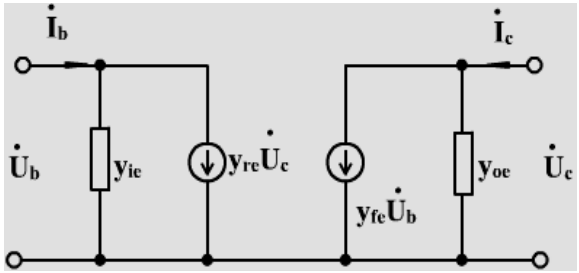
混π参数在一定频率范围内与频率无关,只与工作点有关,因此可用于宽带分析.

2. Y参数等效电路：

在小信号运用时,晶体管可近似为线性网络,因此可用线性方法求解

$$\boxed{\begin{aligned}\dot{I}_b &= y_{ie}\dot{U}_b + y_{re}\dot{U}_c \\ \dot{I}_c &= y_{fe}\dot{U}_b + y_{oe}\dot{U}_c\end{aligned}}$$

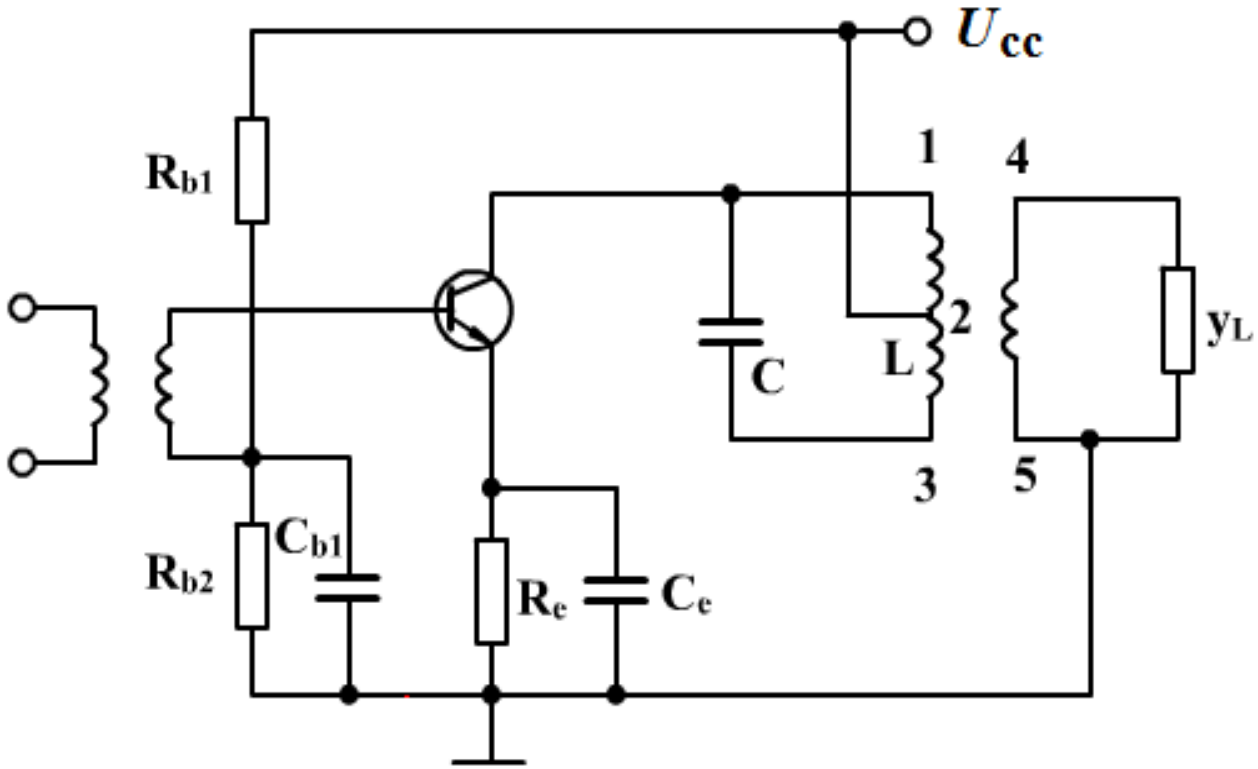




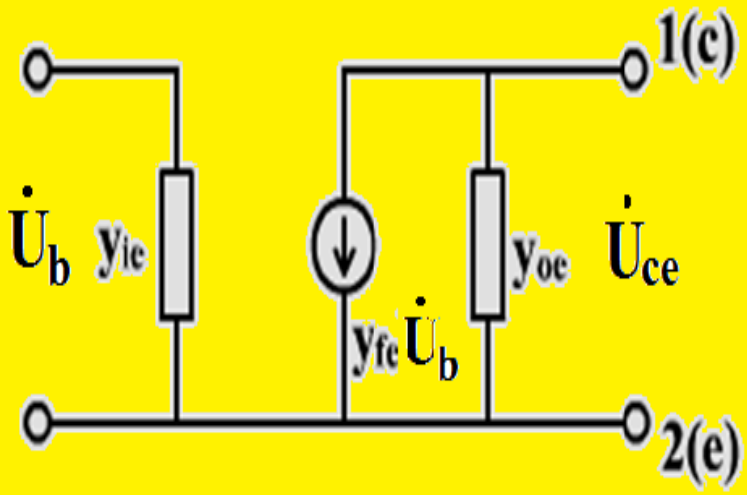
Y参数不仅与工作点有关而且与频率有关,因此只适用于窄带分析。

二、电路组成

(a)图:



(b)图：晶体管等效为Y参数



◆**晶体管接入回路的接入系数：** $p_1 = \dfrac{N_{12}}{N}$

◆**负载端的接入系数：** $p_2 = \dfrac{N_{45}}{N}$负载端

折算到LC回路两端:

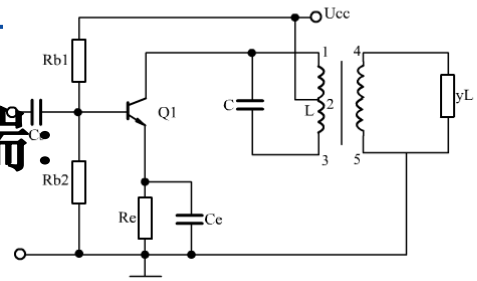
$$g_{oe}' = p_1^2 g_{oe}$$

$$C_{oe}' = p_1^2 C_{oe}$$

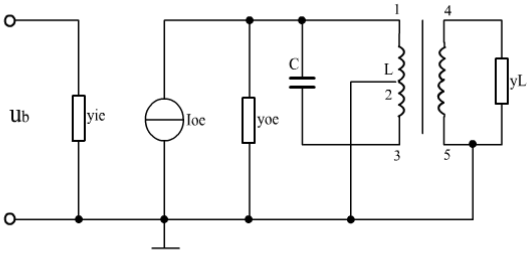
$$\dot{I}_{oe}' = p_1 \dot{I}_{oe} = p_1 y_{fe} \dot{U}_b$$

同理负载端: $g_L' = p_2^2 g_L$

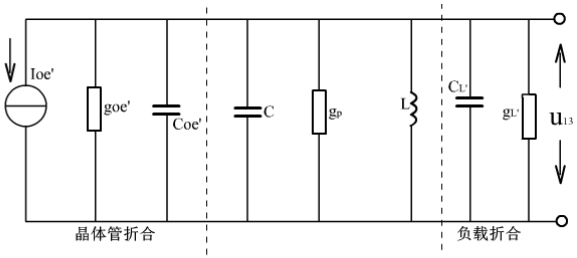
$$c_L' = p_2^2 c_L$$



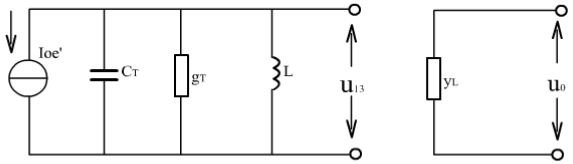
共射放大电路



等效电路(b)



等效电路(c)



等效电路(d)

1-2.exe

最后合并同类元件:

$$C_T = p_1{}^2 C_{oe} + p_2{}^2 C_L + C$$

$$G_T = p_1{}^2 g_{oe} + p_2{}^2 g_L + G_p$$

$$Q_0 = \frac{1}{\omega_0 L G_p}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}} = \frac{1}{2\pi\sqrt{L(P_1^2 C_{oe} + P_2{}^2 C_L + C)}}$$

$$Q_L = \frac{\omega_0 C_T}{G_T} = \frac{1}{\omega_0 L G_T} = \frac{1}{\omega_0 L(P_1^2 g_{oe} + P_2{}^2 g_L + G_p)}$$

三、计算

输出电压：

$$\dot{U}_0 = p_2\dot{U}_{13} = -\frac{p_1 p_2 |y_{fe}| \dot{U}_b}{G_T(1+j\xi)}$$

电压增益：

$$A_V = \frac{\dot{U}_0}{\dot{U}_b} = -\frac{p_1 p_2 |y_{fe}|}{G_T(1+j\xi)}$$

$$A_{V_0} = \frac{-p_1 p_2 |y_{fe}|}{G_T}$$

输出功率：$P_0 = U_0^{\ 2} g_{\mathrm{L}} \quad = \left(\dfrac{p_1 p_2 \left| y_{fe} \right| U_b}{G_{\mathrm{T}}} \right)^2 g_L$

输入功率：$P_{\mathrm{i}} = U_{\mathrm{b}}^{\ 2} g_{\mathrm{ie}}$

功率增益

$$\therefore A_{p0} = \frac{P_0}{P_i} = \frac{\dot{U}_0^{\ 2}}{\dot{U}_b^{\ 2}} \cdot \frac{g_L}{g_{ie}} = A_{V0}^2 \cdot \frac{g_L}{g_{ie}}$$

$$= \left(\frac{p_1 p_2 \left| y_{fe} \right|}{G_T} \right)^2 \cdot \frac{g_L}{g_{ie}}$$

当阻抗匹配时（不考虑回路本身损耗G_p=0）

$$p_1^2 g_{oe} = p_2^2 g_L, \qquad G_T = p_1^2 g_{oe} + p_2^2 g_L = 2p_1^2 g_{oe}$$

$$\therefore A_{po\max} = \frac{(p_1 p_2 |y_{fe}|)^2}{4p_1^4 g_{oe}^2} \times \frac{g_L}{g_{ie}}$$

$$= \frac{p_1^2 p_2^2 |y_{fe}|^2}{4p_1^4 g_{oe}^2} \times \frac{p_1^2 g_{oe}}{p_2^2 g_{ie}} = \frac{|y_{fe}|^2}{4g_{oe} g_{ie}}$$

$$A_{po\max} = \frac{|y_{fe}|^2}{4g_{oe} g_{ie}}$$

功率增益 A_{po} 与最大功率增益 A_{pomax} 之间的关系：

$$A_{po} = \frac{|y_{fe}|^2 p_1^2 p_2^2}{G_T^2} \times \frac{g_L}{g_{ie}} \times \frac{g_{oe}}{g_{oe}} = \frac{|y_{fe}|^2 g_{oe}' g_L'}{G_T^2 g_{ie} g_{oe}}$$

$$= \frac{|y_{fe}|^2}{4 g_{ie} g_{oe}} \times \frac{(g_{oe}' + g_L')^2}{G_T^2} \times \frac{4 g_{oe}' g_L'}{(g_{oe}' + g_L')^2}$$

$$= A_{pomax} \times \eta_1 \times \eta_2$$

其中η_1为回路插入损耗：

$$\eta_1 = \frac{(g_{oe}' + g_L')^2}{G_T^{\ 2}} = \frac{(g_{oe}' + g_L' + g_p - g_p)^2}{G_T^{\ 2}}$$

$$= (1 - \frac{g_p}{G_T})^2 = (1 - \frac{Q_L}{Q_0})^2$$

η_1描述了由于电感线圈损耗不能忽略带来的影响。

$$\eta_1 = (1 - \frac{Q_L}{Q_0})^2 < 1$$

其中η_2为失配功率：

$$\eta_2 = \frac{4g_{oe}'g_L'}{(g_{oe}'+g_L')^2} \times \frac{g_{oe}'^2}{g_{oe}'^2} = \frac{4\frac{g_L'}{g_{oe}'}}{(1+\frac{g_L'}{g_{oe}'})^2} = \frac{4D}{(1+D)^2}$$

其中$D = \frac{g_L'}{g_{oe}'}$称为失配系数

$$\boxed{\eta_2 = \frac{4D}{(1+D)^2} \le 1}$$

阻抗匹配时，p_1、p_2最佳选取:

$$\because p_1^2 g_{\mathrm{oe}} = p_2^2 g_{\mathrm{L}} = \frac{1}{2}(G_{\mathrm{T}} - G_{\mathrm{p}})$$

$$\therefore p_1^{\ 2} = \frac{(G_{\mathrm{T}} - G_{\mathrm{p}})}{2g_{\mathrm{oe}}}$$

$$p_2^{\ 2} = \frac{(G_{\mathrm{T}} - G_{\mathrm{p}})}{2g_{\mathrm{L}}}$$

例4

某中频放大线路如图所示，已知放大器的工作频率为$f_0 = 10.7\text{MHz}$，回路电容$C = 50\text{pF}$，中频变压器接入系数$p_1 = N_1/N = 0.35$，$p_2 = N_2/N = 0.03$，$Q_0 = 100$，晶体管Y参数(在工作频率上)如下：

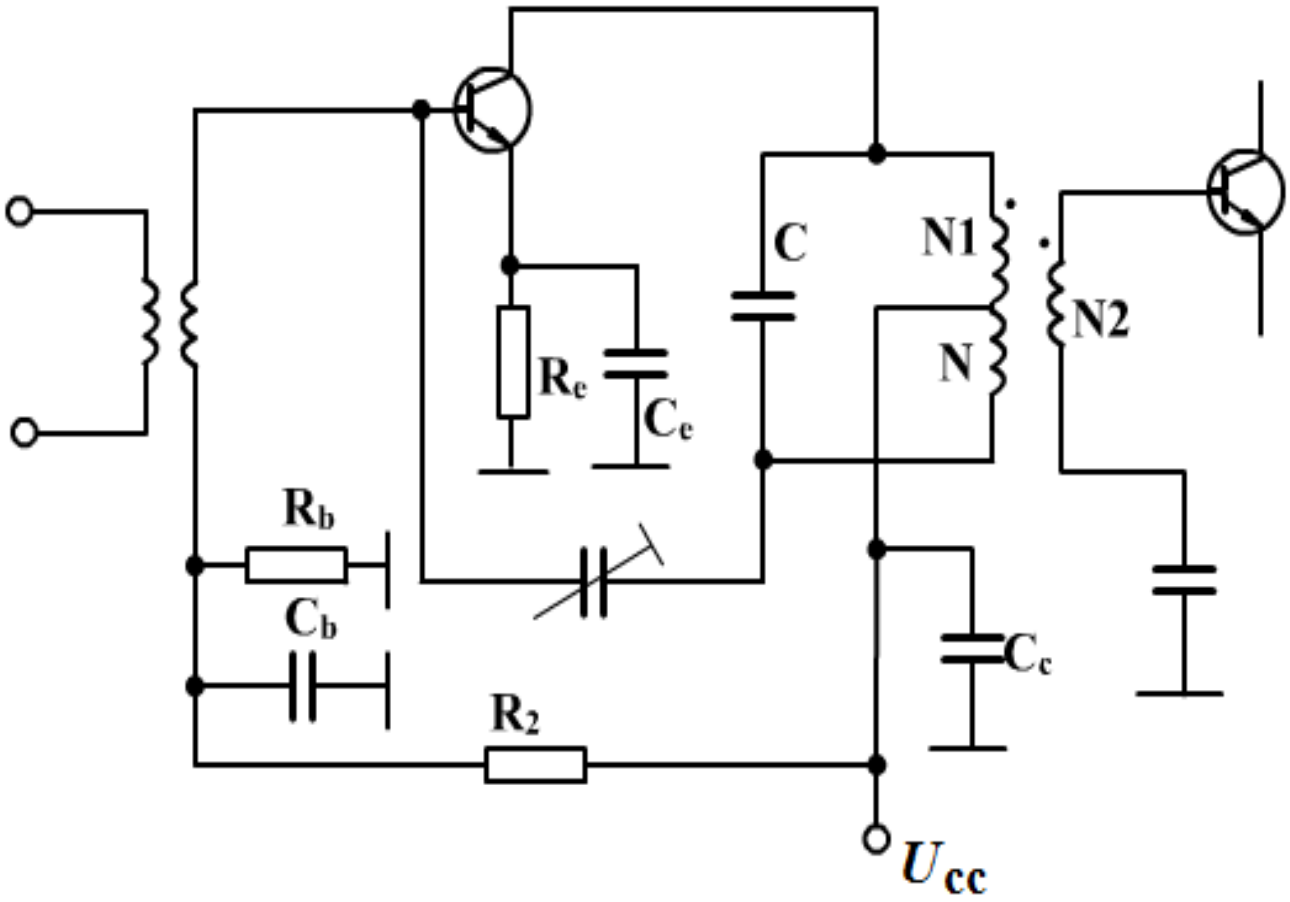
$$g_{ie} = 1.0\text{ms}, C_{ie} = 41\text{pF}, y_{re} = -j180\mu\text{s}.$$

$$y_{fe} = 40\text{ms}, C_{oe} = 4.3\text{pF}, g_{oe} = 45\mu\text{s}$$

设后级输入电导也为g_{ie}，求

(1)回路有载Q_L值和通频带BW

(2)放大器的谐振电压增益.


C
N1
N2
N
Re
Ce
Rb
Cb
R2
Cc
Ucc

要计算$Q_L \rightarrow BW = \dfrac{f_0}{Q_L}$，$\rightarrow Q_L$与有负载阻抗有关，

解

负载有三部分：回路本身负载R_p

$$R_p = \frac{Q_0}{\omega_0 C} = \frac{100}{2\pi \times 10.7 \times 10^6 \times 50 \times 10^{-12}} \approx 30\text{k}\Omega$$

晶体管输出阻抗：

$$R_{oe}' = \frac{1}{p_1^2 g_{oe}} = \frac{1}{0.35^2 \times 45 \times 10^{-6}} = 181\text{k}\Omega.$$

后级输入阻抗：

$$R_{ie}' = \frac{1}{p_2^2 g_{ie}} = \frac{1}{0.03^2 \times 1.0 \times 10^{-3}} = 1.1\text{M}\Omega$$

$$\text{总阻抗}: R_T = R_p // R_{oe}' // R_{ie}' = 30k\Omega // 1.1M\Omega // 181k\Omega$$

$$\approx 25k\Omega$$

$$\text{回路总电容}: C_T = C + C_{ie}' + C_{oe}'$$

$$= C + p_2^2 C_{ie} + p_1^2 C_{oe}$$

$$= 50 + 0.35^2 \times 4.3 + 0.03^2 \times 41 \approx 50.54pf$$

因此f_0基本不变,而

$$Q_L = \omega_0 C_T R_T = 2\pi \times 10.7 \times 10^6 \times 50.54 \times 10^{-12} \times 25 \times 10^3$$

$$= 84.9$$

$$BW = \frac{f_0}{Q_L} = \frac{10.7 \times 10^6}{84.9} = 126kHz$$

谐振时: $f = f_0, \xi = 0$

$$A_{\mathrm{V0}} = -p_1 p_2 \cdot \frac{|y_{\mathrm{fe}}|}{G_T} = -p_1 p_2 |y_{\mathrm{fe}}| \cdot R_{\mathrm{T}}$$

$$= -0.35 \times 0.03 \times 40 \times 10^{-3} \times 25 \times 10^{3}$$

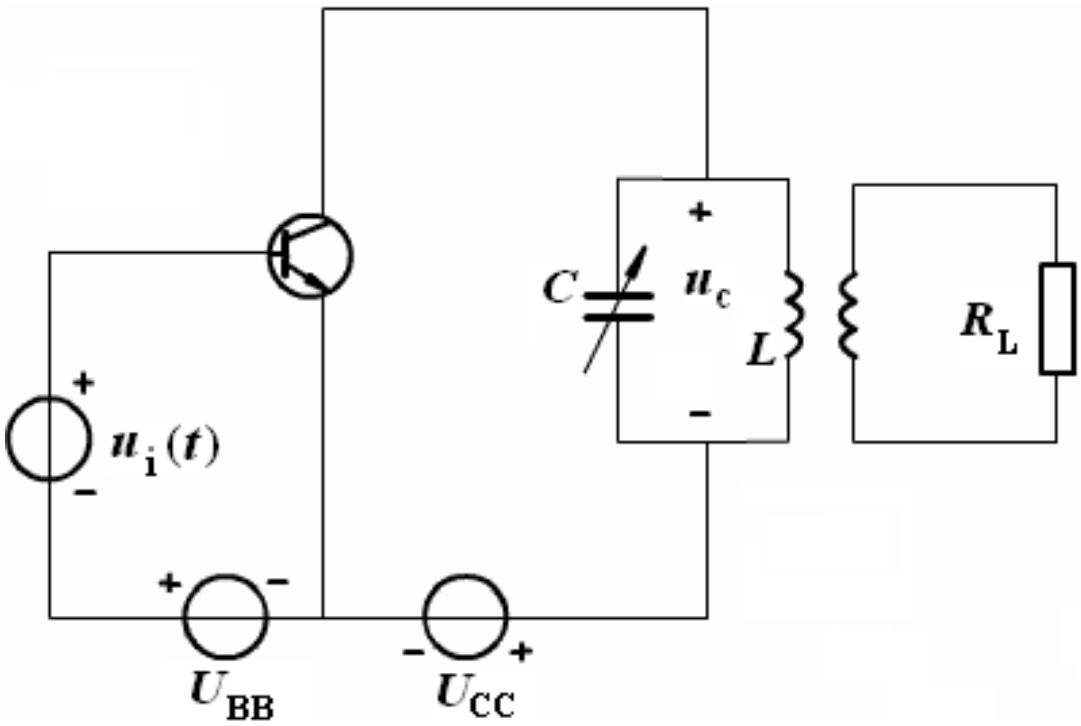
$$= -10.5$$

$$20 \log_{10} A_{V0} = 20.4\mathrm{dB}$$

谐振功率放大器

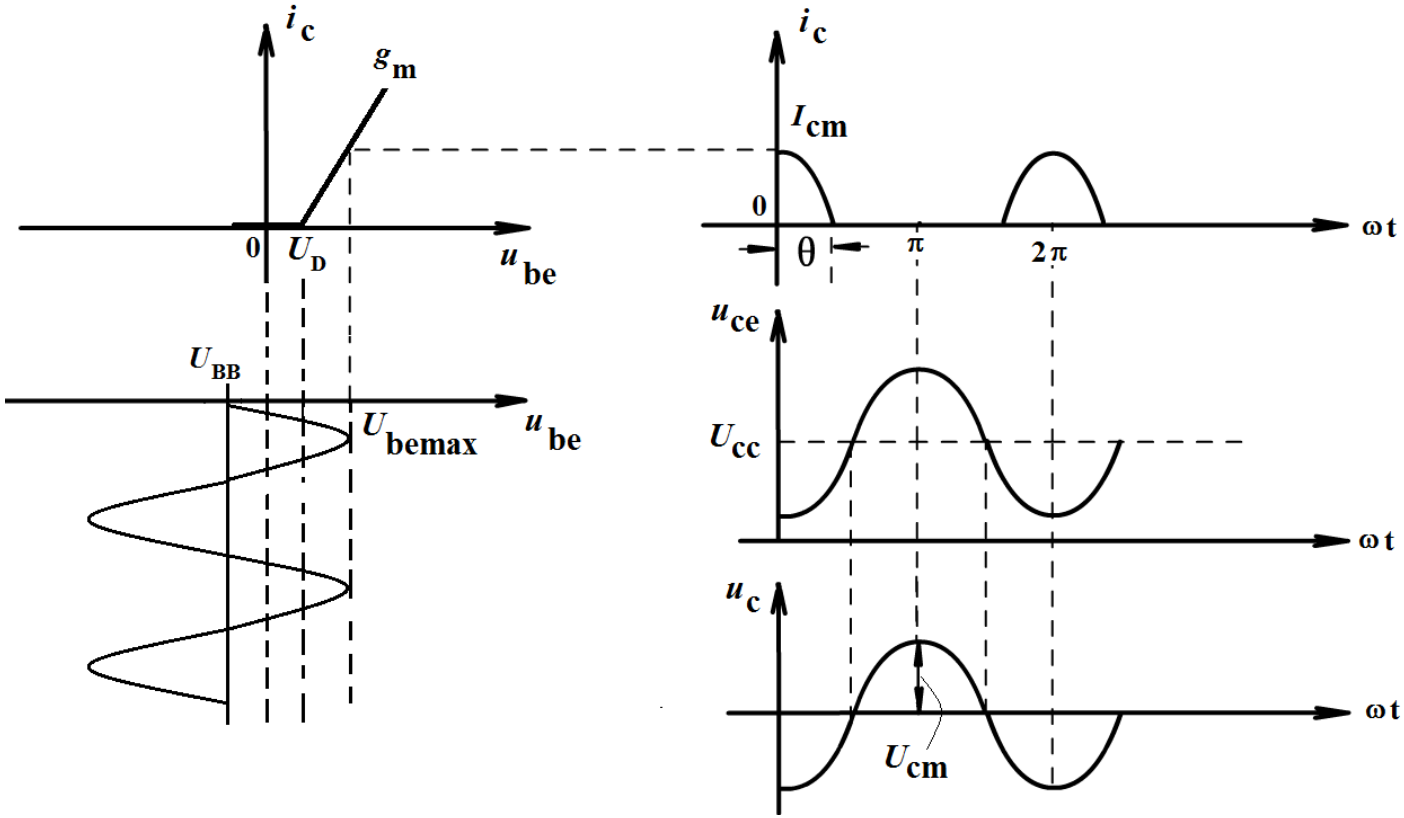
一. 工作原理：

*LC*回路为匹配网络，它们与外接电阻共同组成并联谐振回路。



2-1

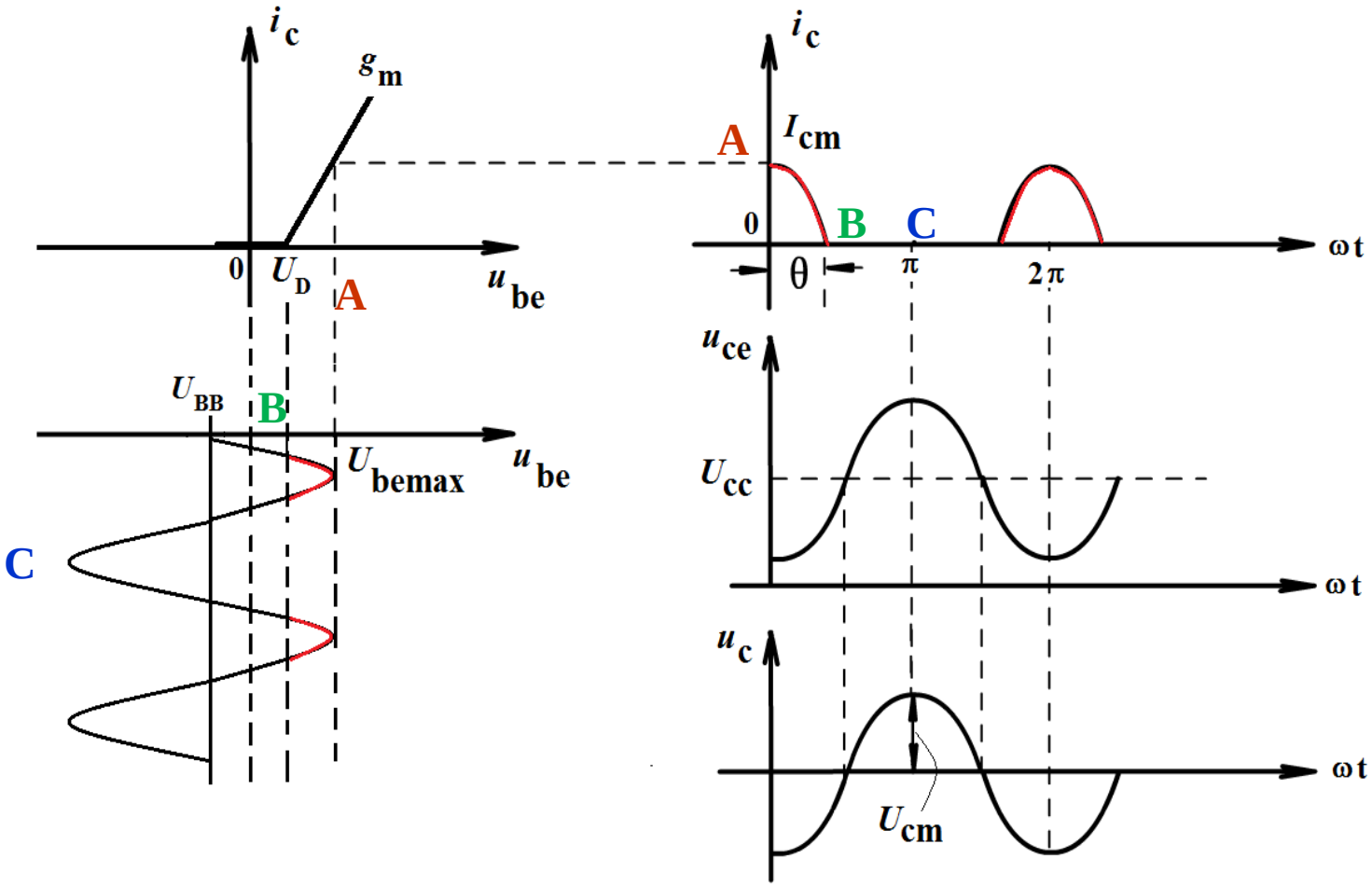
二、折线分析法



为了实现丙类工作，基极偏置电压U_{BB}应设置在功率管的截止区。

折线分析法

集电极电流是一串周期重复的脉冲序列，脉冲宽度小于半个周期。



2-1

基极回路的电压为：

$$u_{be} = U_{BB} + u_i(t)$$

$$= U_{BB} + U_{im}\cos\omega_o t$$

$u_{be} = U_D$**时，** $\cos\theta = \dfrac{U_D - U_{BB}}{U_{im}}$

$U_{BB} \downarrow$，$\theta \downarrow$ 或$U_{im} \downarrow$，$\theta \downarrow$

集电极的电流为余弦脉冲，i_C的傅里叶级数展开:

余弦脉冲的幅值：

$$I_{\mathrm{cm}} = g_{\mathrm{m}}(U_{\mathrm{bema\,x}} - U_{\mathrm{D}}) = g_{\mathrm{m}}(U_{\mathrm{BB}} + U_{\mathrm{im}} - U_{\mathrm{D}})$$

$$= g_{\mathrm{m}} U_{\mathrm{im}}\left(1 - \frac{U_{\mathrm{D}} - U_{\mathrm{BB}}}{U_{\mathrm{im}}}\right) = g_{\mathrm{m}} U_{\mathrm{im}}(1 - \cos\theta)$$

余弦脉冲的表达式：

$$i_c = \begin{cases} g_m(u_{be} - U_D) & u_{be} \geq U_D \\ 0 & u_{be} < U_D \end{cases}$$

$$= \begin{cases} \dfrac{I_{cm}}{1 - \cos\theta}(\cos\omega t - \cos\theta) & u_{be} \geq U_D \\ 0 & u_{be} < U_D \end{cases}$$

$$i_C = I_{C0} + I_{C1}\cos\omega_0 t + I_{C2}\cos 2\omega_0 t + \cdots$$

$$I_{c0} = I_{cm}\frac{\sin\theta - \theta\cos\theta}{\pi(1-\cos\theta)} = I_{cm}\alpha_0(\theta)$$

$$I_{c1} = I_{cm}\frac{\theta - \sin\theta\cos\theta}{\pi(1-\cos\theta)} = I_{cm}\alpha_1(\theta)$$

$$\vdots \qquad \vdots$$

$$I_{cn} = I_{cm}\frac{2\sin n\theta - 2n\sin\theta\cos n\theta}{n\pi(n^2-1)(1-\cos\theta)} = I_{cm}\alpha_n(\theta)$$

$\alpha_n(\theta)$ —— **n次谐波的分解系数**

三. 性能分析与计算(近似分析)

假设：谐振回路具有理想滤波特性。只产生基波电压，其他分量忽略不计。

$$u_{be} = U_{BB} + U_{im}\cos\omega t$$

$$u_{ce} = U_{CC} - U_{cm}\cos\omega t$$

谐振电阻： $$R_p = \frac{U_{cm}}{I_{c1}}$$

直流功率： $P_{dc} = U_{cc} I_{c0} = \alpha_0 I_{cm} U_{cc}$

输出功率：

$$P_o = \frac{1}{2} U_{cm} I_{c1} = \frac{1}{2} \alpha_1 U_{cm} I_{cm} = \frac{1}{2} I_{c1}^2 R_p$$

其中 $I_{c0} = \alpha_0 I_{cm}$ $I_{c1} = \alpha_1 I_{cm}$

损耗功率： $P_c = P_{dc} - P_o$

集电极效率： $\eta_c = \dfrac{P_o}{P_{dc}}$

令集电极电源电压利用系数：

$$P_o = \frac{1}{2}\xi U_{CC}\alpha_1 I_{cm} \qquad \xi = \frac{U_{cm}}{U_{CC}}$$

$$\eta_c = \frac{\frac{1}{2}\xi U_{cc}\alpha_1 I_{cm}}{U_{cc}\alpha_0 I_{cm}} = \frac{1}{2}\xi\frac{\alpha_1}{\alpha_0} = \frac{1}{2}\xi g_1$$

因此导通角 θ **决定了** η_c，P_o

$\theta \approx 70^\circ \sim 80^\circ$时，$\eta_c$，$P_o$ 较大。

四、工作状态

假设一：谐振回路具有理想滤波特性，其上只能产生基波分量；

假设二：功率管的特性用输入和输出静态特性曲线表示，其高频效应可忽略。

2-2

动态线（Dynimic Line）：

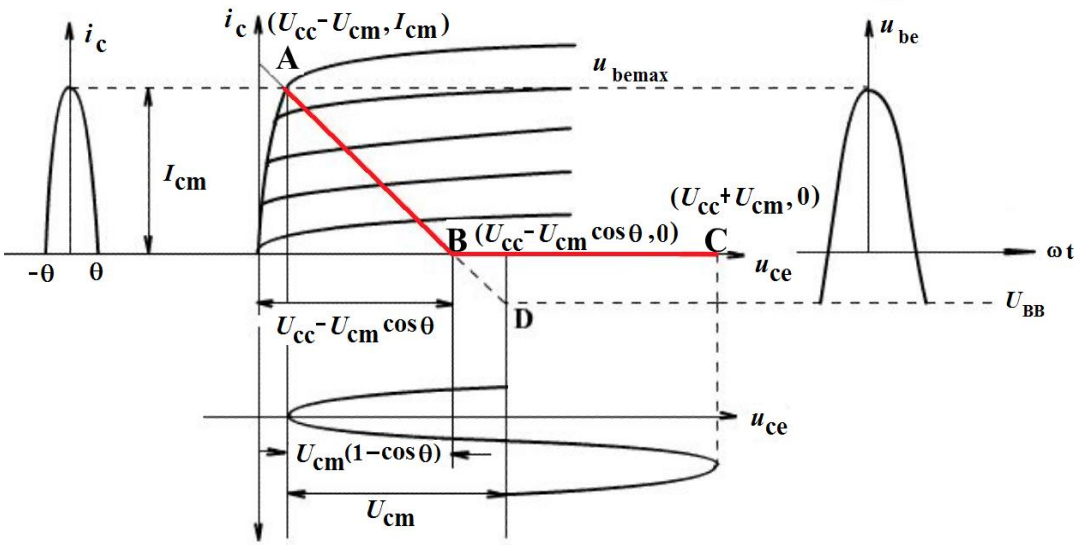
根据u_{be}和u_{ce}的值在以u_{be}为参变量的输出特性曲线上找出对应的动态点和由此确定的i_c值并画出i_c的波形。其中，动态点的连线称为谐振功率放大器的动态线。

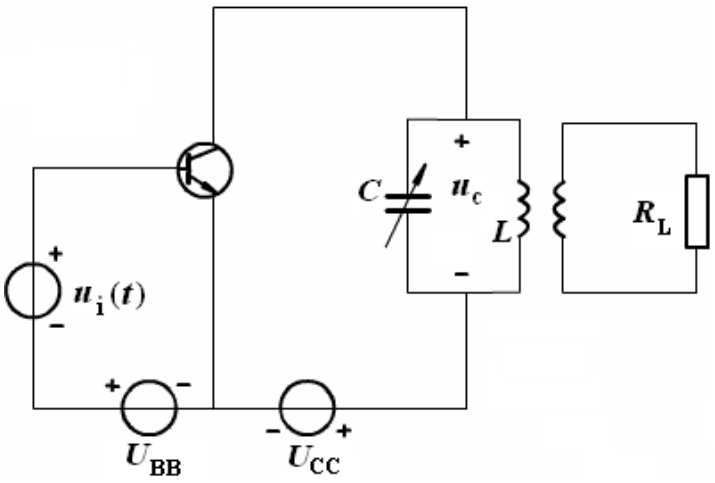
A点： $\omega t = 0$ ， $u_{BE} = U_{BB} + U_{im}$ ， $u_{CE} = U_{CC} - U_{cm}$

B点： $\omega t = \theta$ ， $i_c = 0$ ， $u_{CE} = U_{CC} - U_{cm}\cos\theta$

C点： $\omega t = \pi$ ， $i_c = 0$ ， $u_{CE} = U_{CC} + U_{cm}$

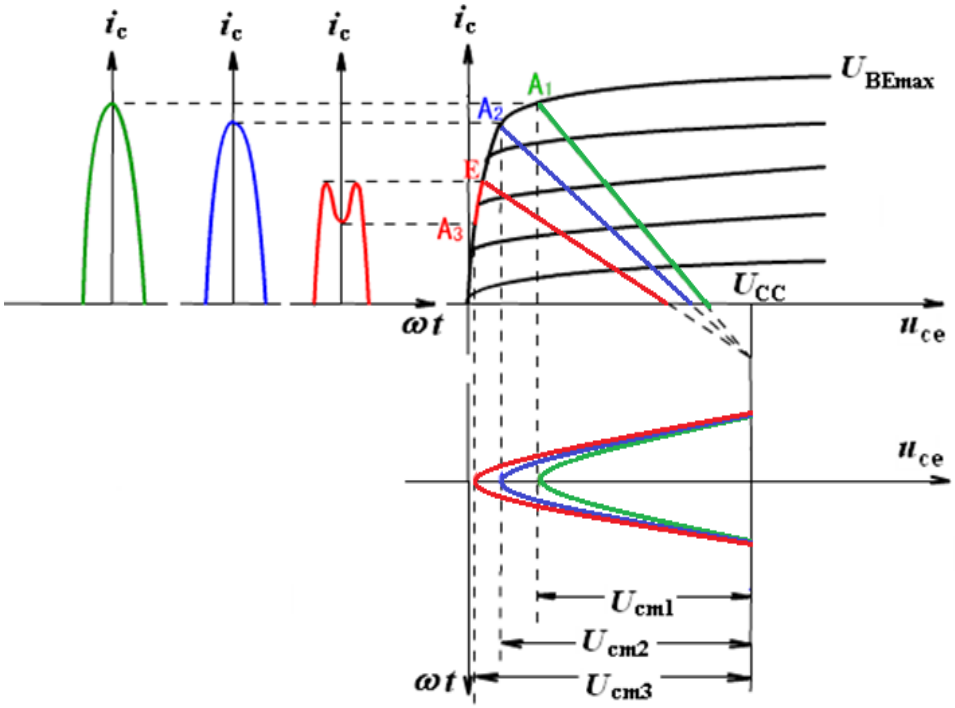
2-2





直线AB、BC动态特性曲线

- 当A_2点处于临界区称为临界（Critical）状态
- 当A_1点处于放大区称为欠压（Undervoltage）状态
- 当A_3点处于饱和区称为过压（Overvoltage）状态



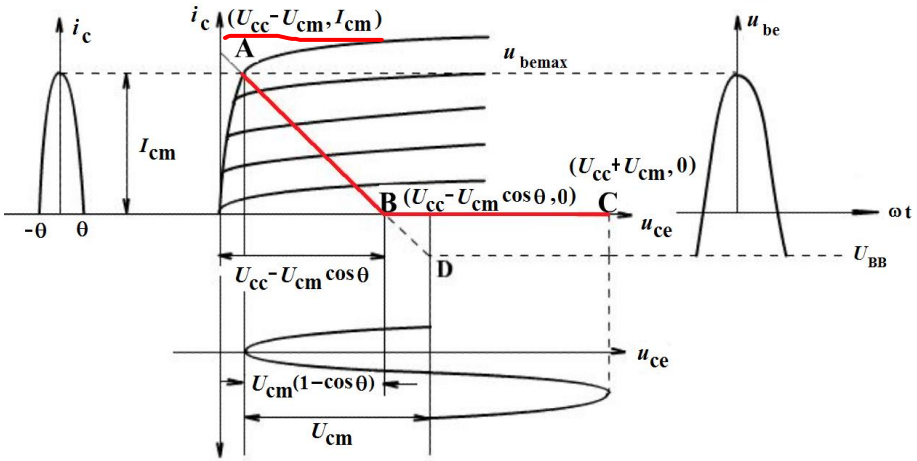
利用A，B两点直线计算斜率：

$$A\ (U_{CE\min}\ ,\ I_{cm})\ ,\ B(U_{CC}-U_{cm}\cos\theta\ ,\ 0)$$

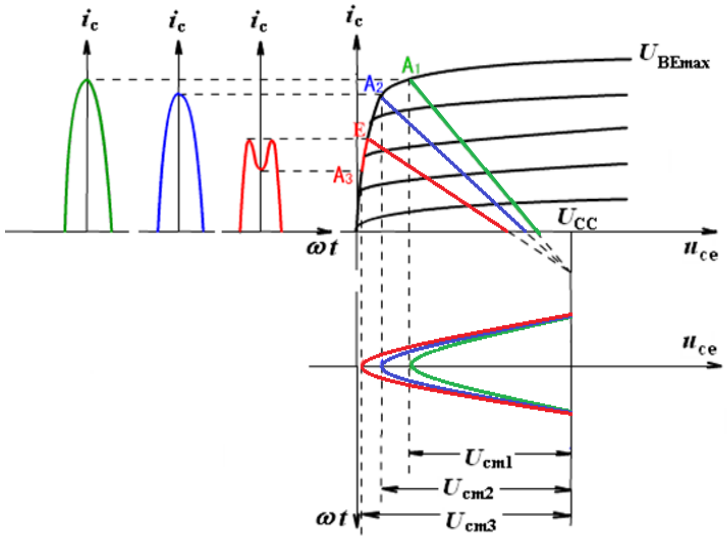
在A没有进入饱和区时，动态特性曲线斜率为

$$k=-\frac{I_{cm}}{U_{cm}(1-\cos\theta)}=-\frac{I_{cm}}{I_{c1}R_p(1-\cos\theta)}$$

$$=-\frac{1}{\alpha_1 R_p(1-\cos\theta)}$$



2-2



2-3

1. A_1在欠压区，i_c为余弦脉冲，脉冲高度随U_{cm}增大略有减小。
2. A_2临界状态。
3. A_3过压区，i_c为中间凹陷的脉冲波，随着U_{cm}增大脉冲波的凹陷加深，高度减小。

思考题： 在过压区为什么会出现凹陷?

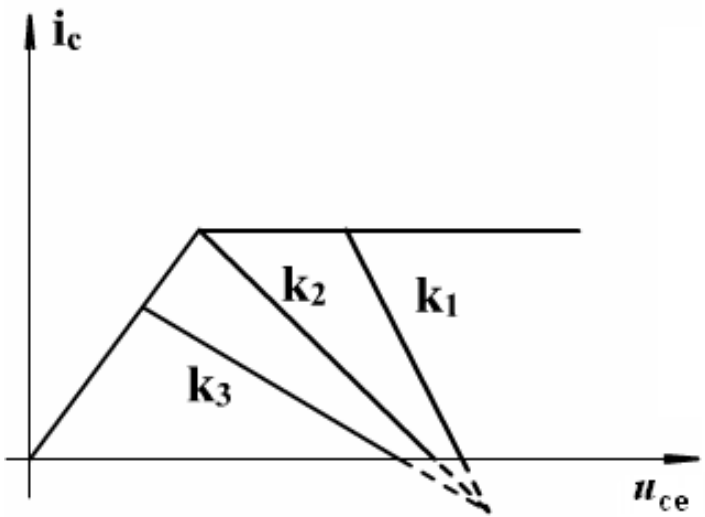
当u_{BE}向U_{BEmax}增大，u_{CE}向着U_{CEmin}减小时，对应的动态点先到达临界点(图中E点)，i_c值最大，而后进入饱和区，i_c值减小，出现凹陷。直到A_3点，i_c值达到最小。

五. R_p ，U_{CC} ，U_{im} ，U_{BB} 对性能影响的定性分析

1. 负载特性：

U_{CC} ，U_{im} ，U_{BB} **一定，放大器性能随 R_p 变化的特性。**

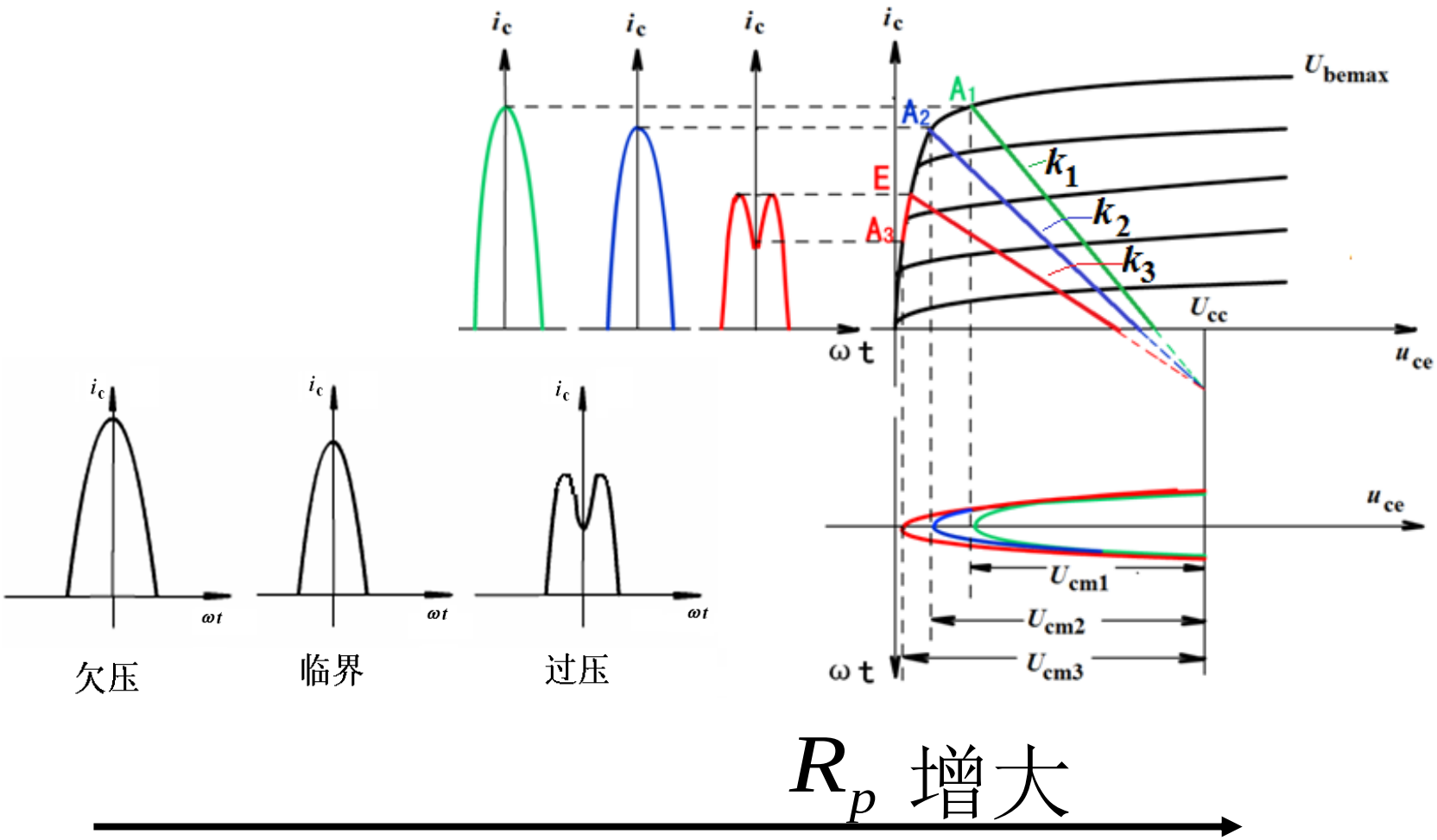
R_p **改变，相当于改变动态特性的斜率 k**



$$|k| = \frac{1}{\alpha_1(1-\cos\theta)R_p}$$

$$|k_3| < |k_2| < |k_1|$$

$$R_{p3} > R_{p2} > R_{p1}$$



R_p 增大

集电极电流由余弦脉冲变化为中间凹陷的脉冲波。

在欠压区

$$U_{cm} = I_{c1} R_p$$

$$P_o = \frac{I_{c1}^2}{2} R_p$$

与 R_p 近似成正比关系

(在欠压区)

$$P_D = U_{CC} I_{c0}$$

在欠压区与 I_{c0} 一样略有减小

$$\eta_c = \frac{P_o}{P_D}$$

$$\Rightarrow \quad P_C$$

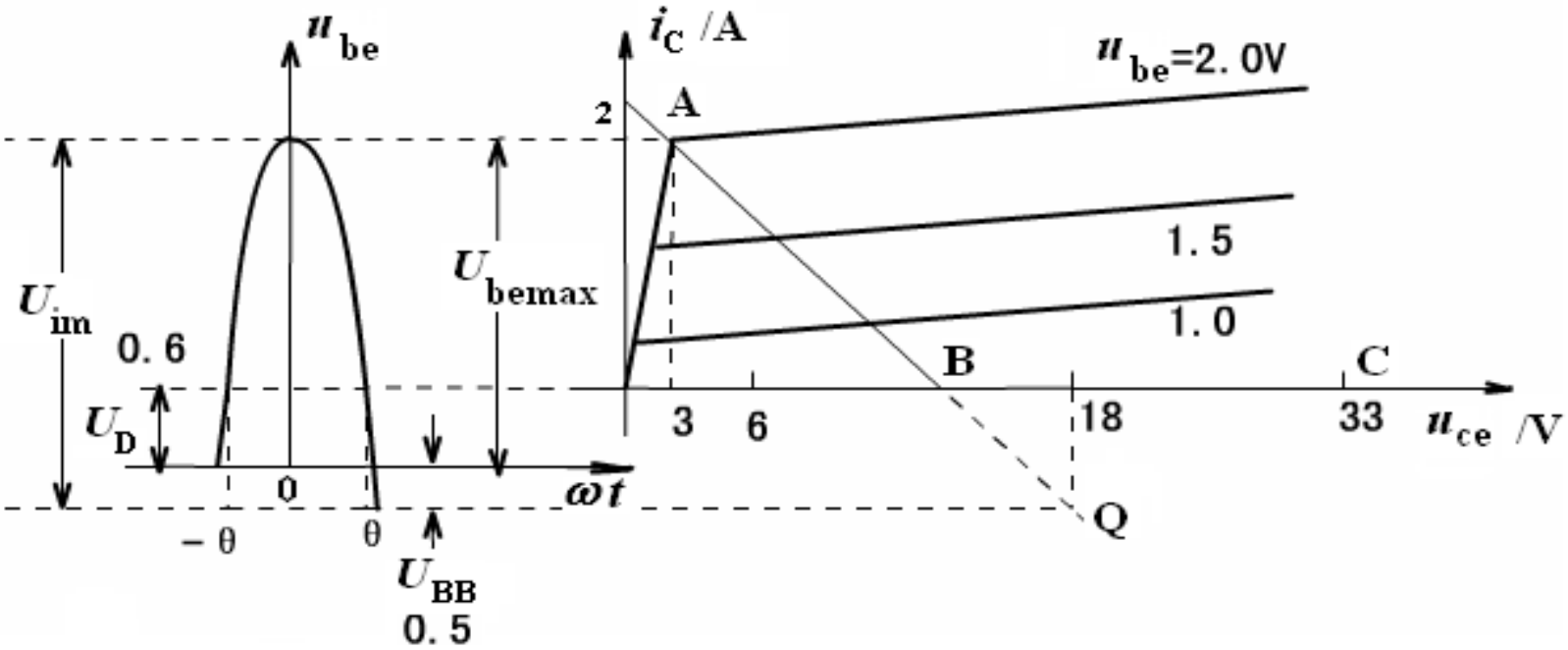
在过压区

- $R_p \nearrow$ i_c 出现凹陷。I_{c0}，$I_{c1} \searrow$ 而 U_{cm} 略有增加
- P_o，$P_D \searrow$ 且 P_o 比 P_D 减小得慢。从而 η_c 略有增加
- P_C 略有减小。

在临界点 P_0 最大，η_c 较大，放大器接近最佳性能，

例题

某谐振频率放大器的动特性如图所示，试回答并计算以下各选项： (1)此时功放工作于何种状态？写出A，B，C三点坐标,画出i_c的波形。 (2)计算 θ ，P_D ，P_C 和 R_p 。 (3)若要求功率放大器的效率最大，应如何调整?



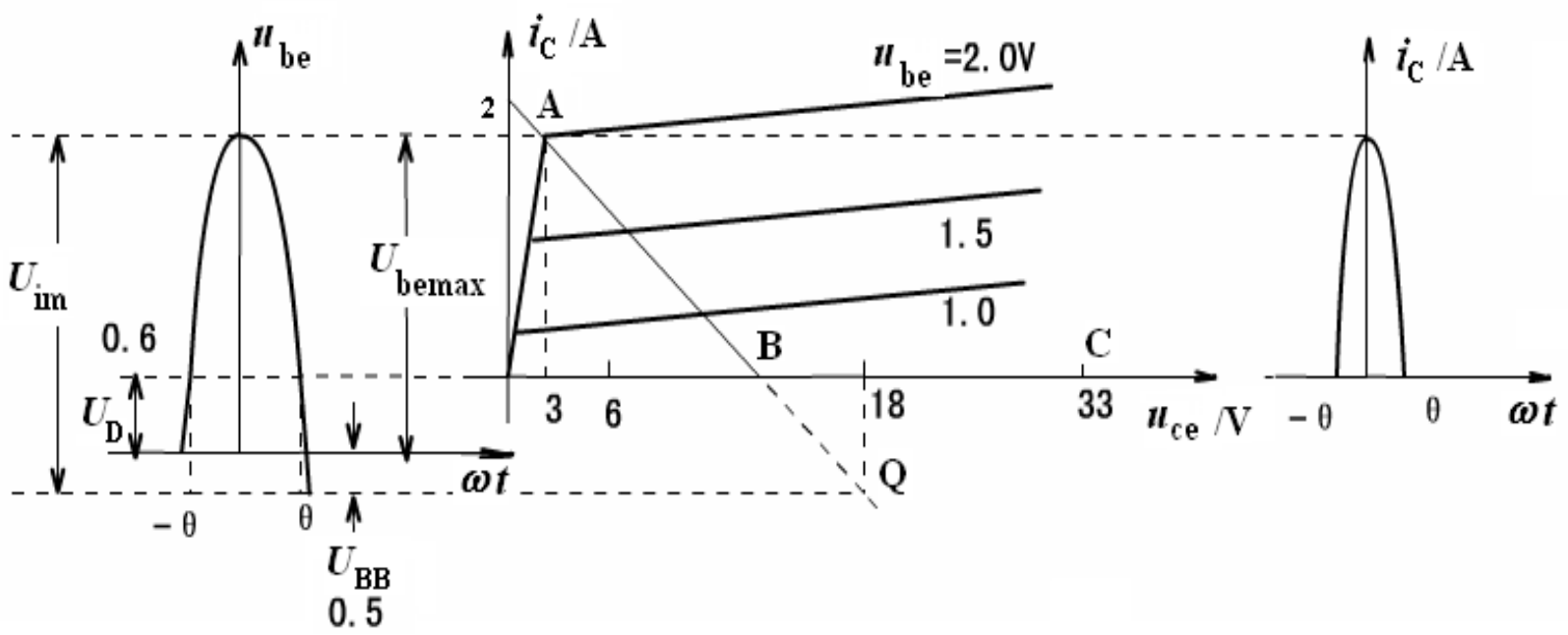
解

(1) A，B，C三点坐标：

$$A(U_{CC}-U_{cm}\ ,I_{cm})\quad B(U_{CC}-U_{cm}\cos\theta\ ,0)\quad C(U_{CC}+U_{cm}\ ,0)$$

ABC动态线：由动态线知正好达到临界饱和线，

故工作在临界状态，i_c的波形如下。



(2) **由图可知：** $U_{BB}=-0.5\ \text{V},\quad U_D=0.6\ \text{V},$

$U_{CC}=18\ \text{V}\qquad I_{cm}=2.0\ \text{A},$

$$\cos\theta=\frac{U_D-(U_{BB})}{U_{im}}=\frac{0.6-(-0.5)}{2-(-0.5)}=0.44$$

$$\therefore\theta=64^\circ$$

由 $u_{CE\max}$ **和** $u_{CE\min}$ **可求得**

$$U_{cm}=\frac{U_{CE\max}-U_{CE\min}}{2}=15\ \text{V}$$

$$P_o = \frac{I_{c1}U_{cm}}{2} = \frac{\alpha_1 I_{cm}U_{cm}}{2} = \frac{2\times 0.410\times 15}{2} = 6.15\ \text{W}$$

$$P_D = U_{CC}I_{c0} = \alpha_0 I_{cm}U_{CC} = 18\times 2\times 0.232 = 8.35\ \text{W}$$

$$P_C = P_D - P_o = 8.35 - 6.15 = 2.2\ \text{W}$$

$$\eta_c = \frac{P_o}{P_D} = \frac{6.15}{8.35} = 74\%$$

$$R_p = \frac{U_{cm}}{I_{c1}} = \frac{U_{cm}}{\alpha_1 I_{cm}} = \frac{15}{0.410\times 2} = 18.29\ \Omega$$

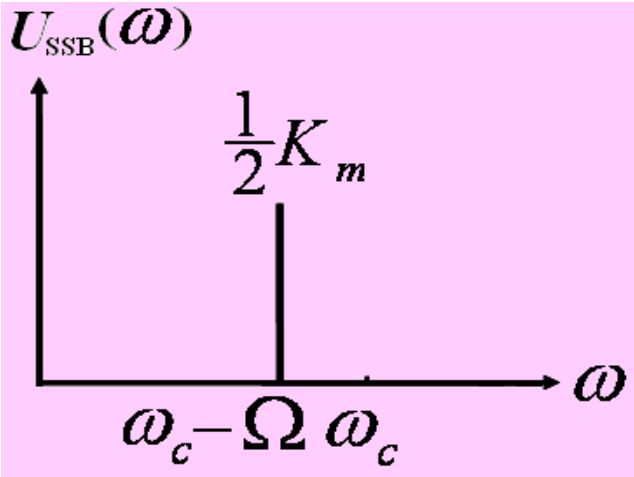
(3) 弱过压状态效率最大（但输出功率下降）

作业：2-5，2-6 的类型题重点练习

幅度的调制和解调

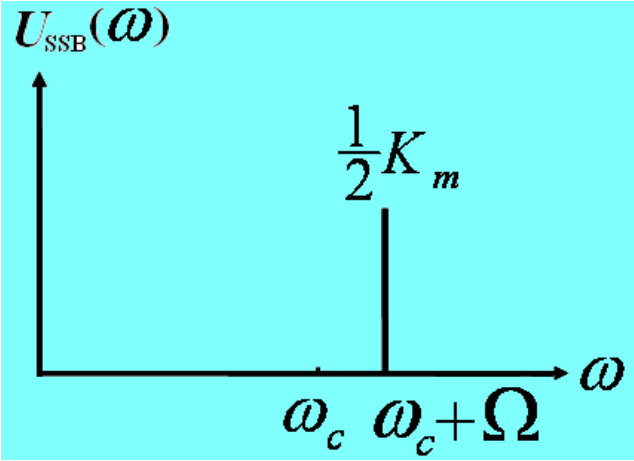
三、抑制载波的单边带调制

(Single Sideband Modulation. SSB)



下边带

$$u_{SSB}=\frac{K_m}{2}\cos(\omega_c-\Omega)t$$

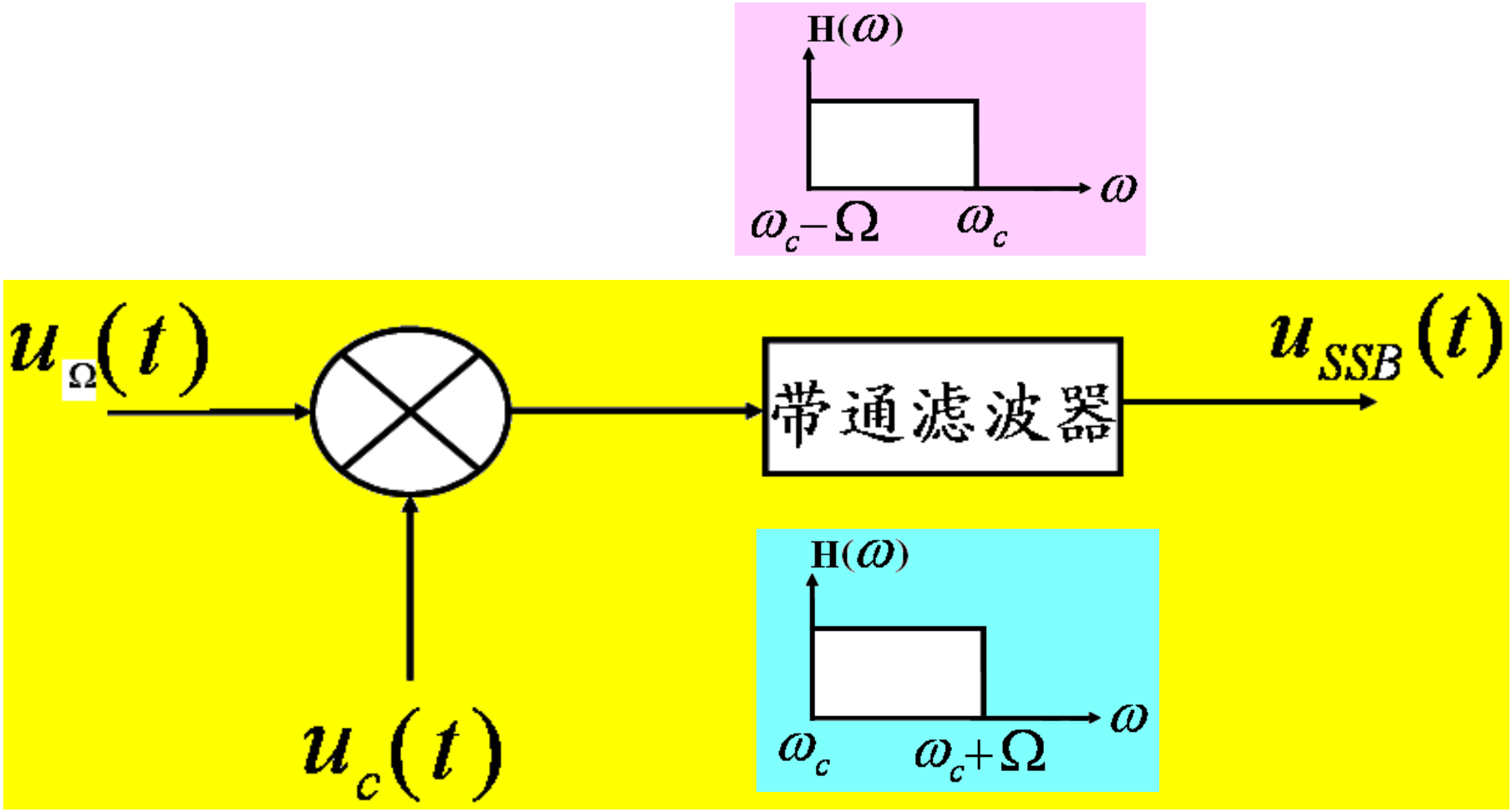


上边带

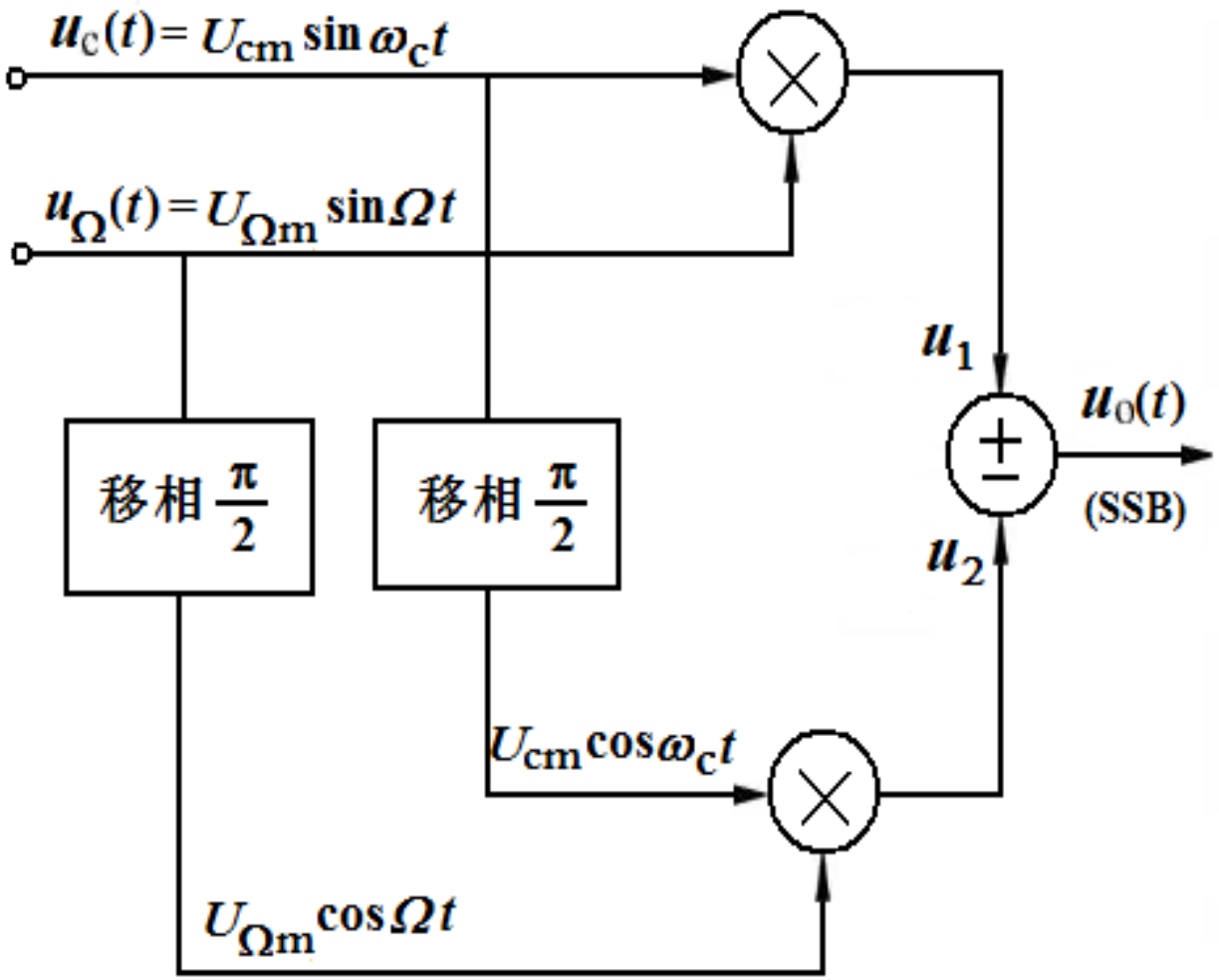
$$u_{SSB}=\frac{K_m}{2}\cos(\omega_c+\Omega)t$$

$$BW_{SSB}=\Omega$$

（1） 滤波法



（２） 移相法

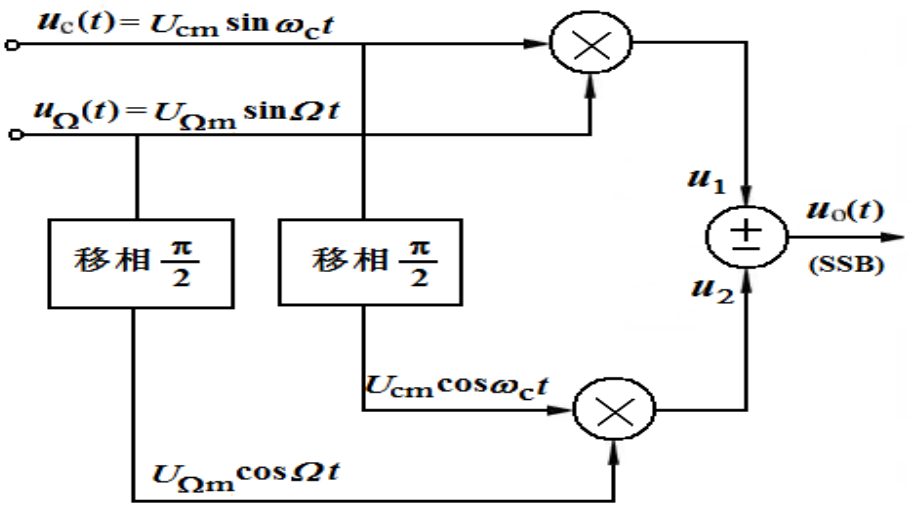

$u_c(t) = U_{cm}\sin\omega_c t$
$u_\Omega(t) = U_{\Omega m}\sin\Omega t$
移相 $\frac{\pi}{2}$
移相 $\frac{\pi}{2}$
u_1
u_2
$u_0(t)$
(SSB)
$U_{cm}\cos\omega_c t$
$U_{\Omega m}\cos\Omega t$

$$u_1(t) = A_m U_{\Omega m} U_{cm} \sin \Omega t \sin \omega_c t$$

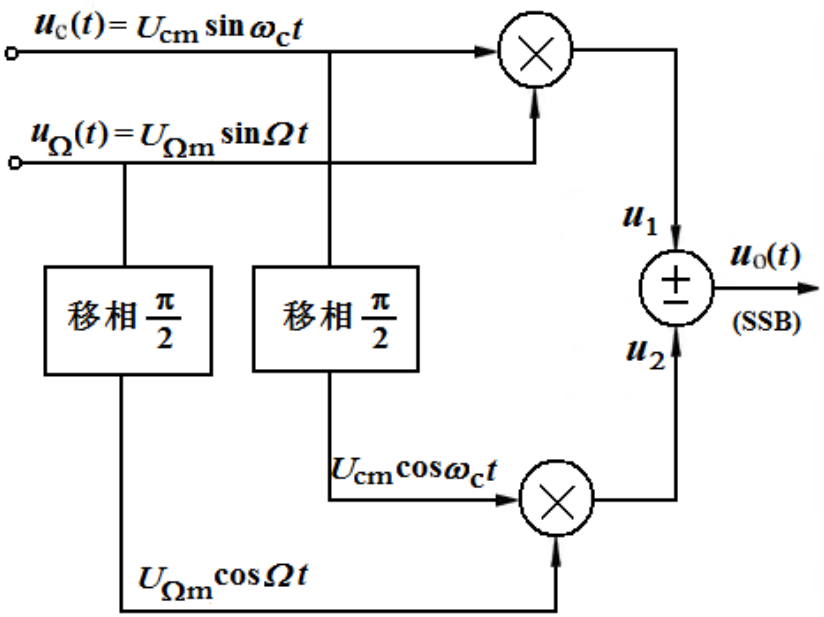
$$= \frac{1}{2} A_m U_{\Omega m} U_{cm} \left[\cos(\omega_c - \Omega)t - \cos(\omega_c + \Omega)t \right]$$

$$u_2(t) = A_m U_{\Omega m} U_{cm} \cos \Omega t \cos \omega_c t$$

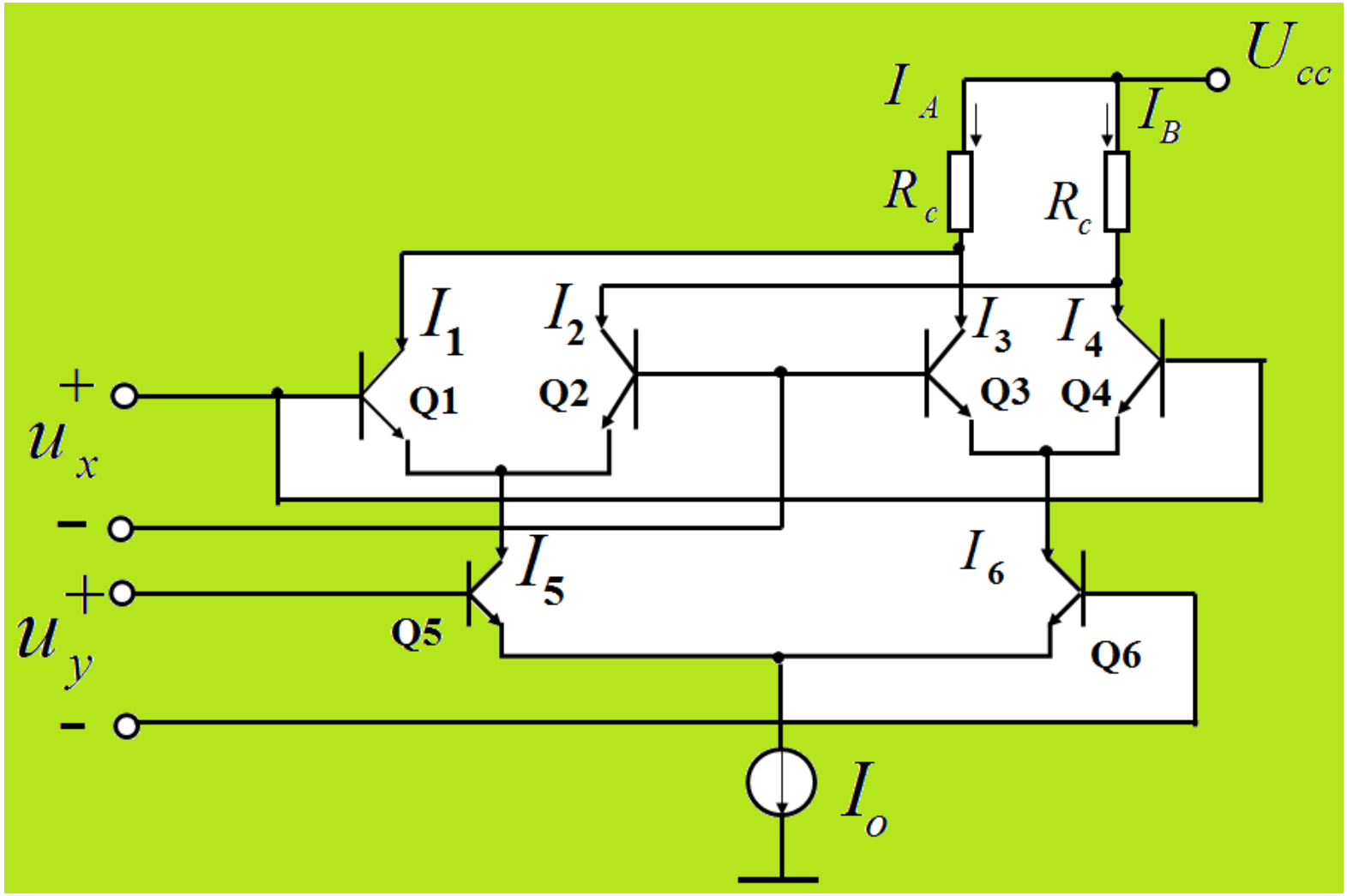
$$= \frac{1}{2} A_m U_{\Omega m} U_{cm} \left[\cos(\omega_c - \Omega)t + \cos(\omega_c + \Omega)t \right]$$



$$u_{SSB}(t)=\begin{cases} u_1(t)-u_2(t)=A_m U_{\Omega m} U_{cm}\cos(\omega_c+\Omega)t \\ u_1(t)+u_2(t)=A_m U_{\Omega m} U_{cm}\cos(\omega_c-\Omega)t \end{cases}$$



2. 双差分对乘法器（Gilbert）



双差分对乘法器

$$I_5 - I_6 = I_o \tanh(\frac{u_y}{2U_T})$$

$$I_1 - I_2 = I_5 \tanh(\frac{u_x}{2U_T})$$

$$I_4 - I_3 = I_6 \tanh(\frac{u_x}{2U_T})$$

双差分对乘法器

$$i = I_A - I_B = I_1 - I_2 - (I_4 - I_3)$$

$$= (I_5 - I_6)\tanh(\frac{u_x}{2U_T}) = I_o \tanh(\frac{u_y}{2U_T})\tanh(\frac{u_x}{2U_T})$$

$$\therefore u_o = iR_c = I_o R_c \tanh(\frac{u_y}{2U_T})\tanh(\frac{u_x}{2U_T})$$

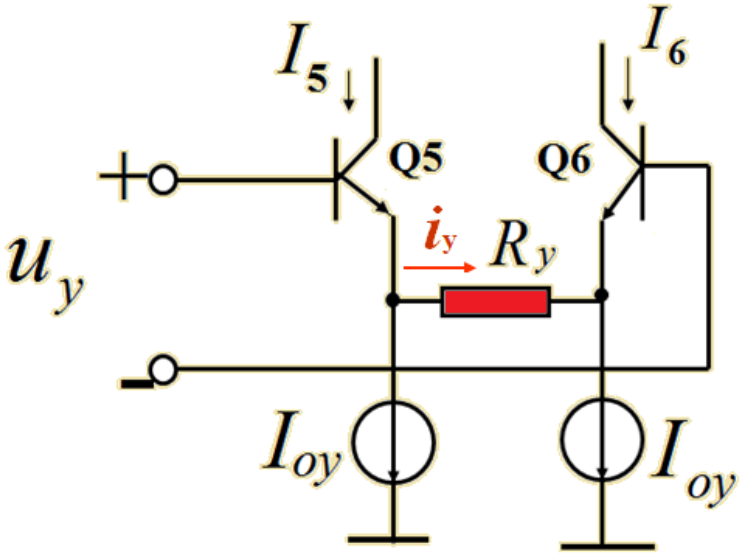
(1) 当 $u_x < U_T$ 时 $u_y < U_T$

$$u_o = \frac{I_o R_c}{4U_T^2} u_x u_y = A u_x u_y$$

(2) 当 $|u_y| < 26\ \text{mV}$ $|u_x| > 260\ \text{mV}$

$$u_o = \frac{I_o R_c}{2U_T} u_y S_2(\omega_1 t)$$

3.扩展 u_y 的动态范围



$$\left.\begin{aligned} R_y &>> 2r_{e5} = \frac{2U_T}{I_{oy}} \\ R_y &>> 2r_{e6} = \frac{2U_T}{I_{oy}} \end{aligned}\right\}$$ 共射极结电阻

$$u_y = u_{BE5} - u_{BE6} + i_y R_y$$

3.扩展 u_y 的动态范围

$$u_y = u_{BE5} - u_{BE6} + i_y R_y$$

其中 $u_{BE5} - u_{BE6} = U_T \ln(\frac{I_5}{I_6})$

$$\therefore \quad u_y = U_T \ln(\frac{I_5}{I_6}) + i_y R_y$$

$$I_6 \approx I_{e6} = I_{oy} - i_y = (1 - \frac{i_y}{I_{oy}}) I_{oy}$$

$$I_5 \approx I_{e5} = I_{oy} + i_y = (1 + \frac{i_y}{I_{oy}}) I_{oy}$$

3.扩展 u_y 的动态范围

$$u_y = U_T \ln \frac{1 + i_y / I_{oy}}{1 - i_y / I_{oy}} + i_y R_y$$

$$= U_T \left[\ln(1 + i_y / I_{oy}) - \ln(1 - i_y / I_{oy}) \right] + i_y R_y$$

3.扩展 u_y 的动态范围

由 $$\ln(1+x)=x-\frac{1}{2}x^2+\frac{1}{3}x^3-\frac{1}{4}x^4+\cdots$$

$$\ln(1-x)=-x-\frac{1}{2}x^2-\frac{1}{3}x^3-\frac{1}{4}x^4-\cdots$$

当 $\dfrac{i_y}{I_{oy}}=x<0.5$ 时，忽略三次方以上项

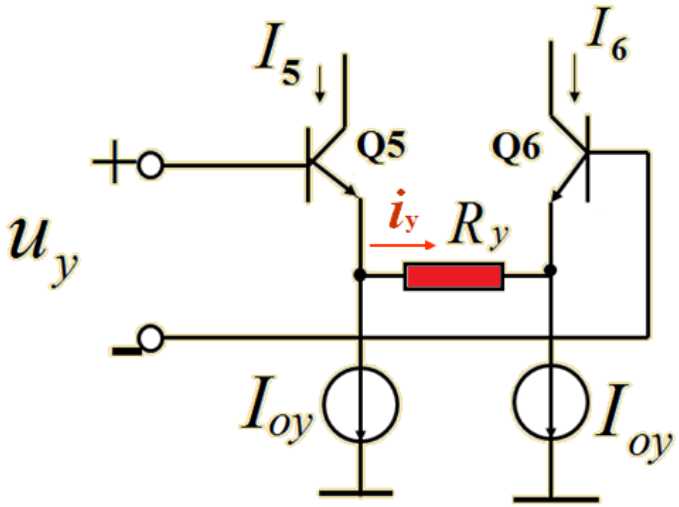
3.扩展 u_y 的动态范围

$$u_y \approx U_T \frac{2i_y}{I_{oy}} + i_y R_y$$

（其中 $\frac{2U_T}{I_{oy}} = 2r_{e5} << R_y$）

$$= \frac{2U_T}{I_{oy}} i_y + i_y R_y \approx i_y R_y$$

$$I_5 - I_6 = 2i_y = \frac{2u_y}{R_y}$$



3.扩展 u_y 的动态范围

$$\therefore \quad i = I_A - I_B = (I_5 - I_6)\tanh(\frac{u_x}{2U_T})$$

$$= \frac{2u_y}{R_y}\tanh(\frac{u_x}{2U_T})$$

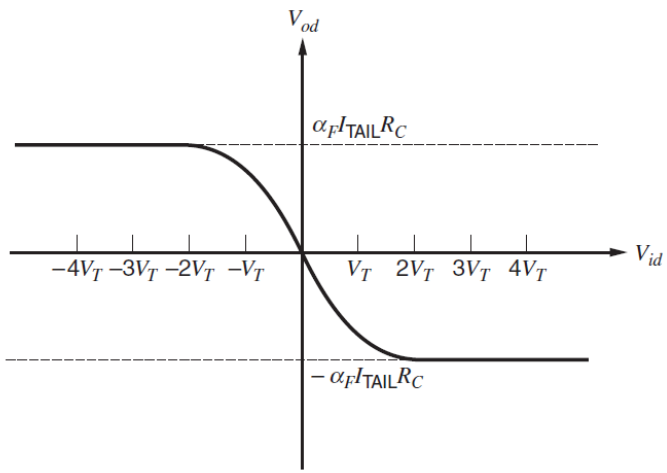
$$u_o = R_c \times i = \frac{2R_c}{R_y}u_y \tanh(\frac{u_x}{2U_T})$$

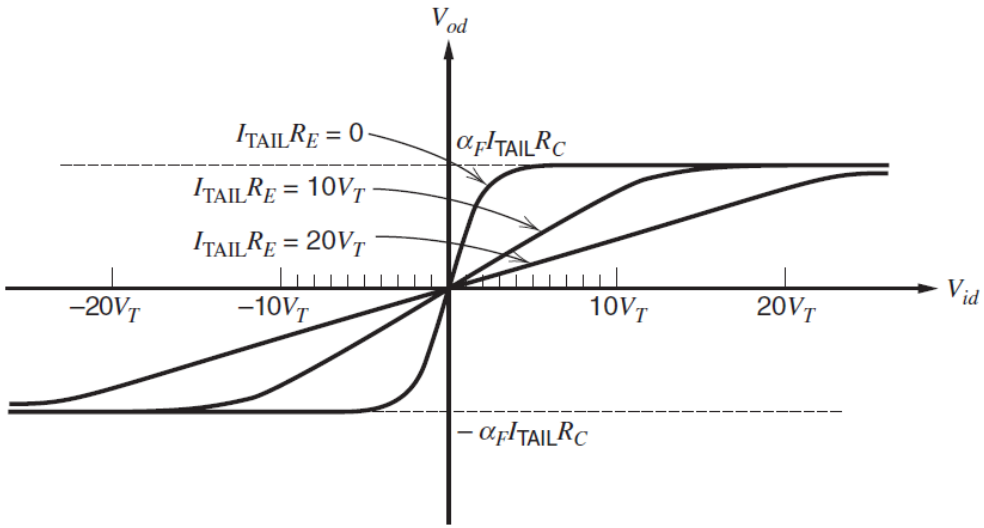
(1) u_x 为小信号

$$u_o = \frac{R_e}{R_y U_T} u_x u_y = A u_x u_y$$

(2) u_x 为大信号

$$u_o = \frac{2R_c}{U_T} u_y S_2(\omega_1 t)$$





3.扩展 u_y 的动态范围

u_y 的动态范围（在 $I_{oy} = 0.5\,\text{mA}$，$R_y = 1\,\text{k}\Omega$ 时）

$$(-276\,\text{mV}\ ,\ +276\,\text{mV})$$

比不接 R_y 时扩大了10倍

集成电路MC1496/MC1596 (XFC1596 , F1596)

MC1596

双差分对乘法器

注意：双差分对乘法器可以有多种用途，比如调制，检波，混频等等。根据不同的题目条件，比如输入信号的情况，要判断该电路在该种条件下的具体功能，然后再对应解题。

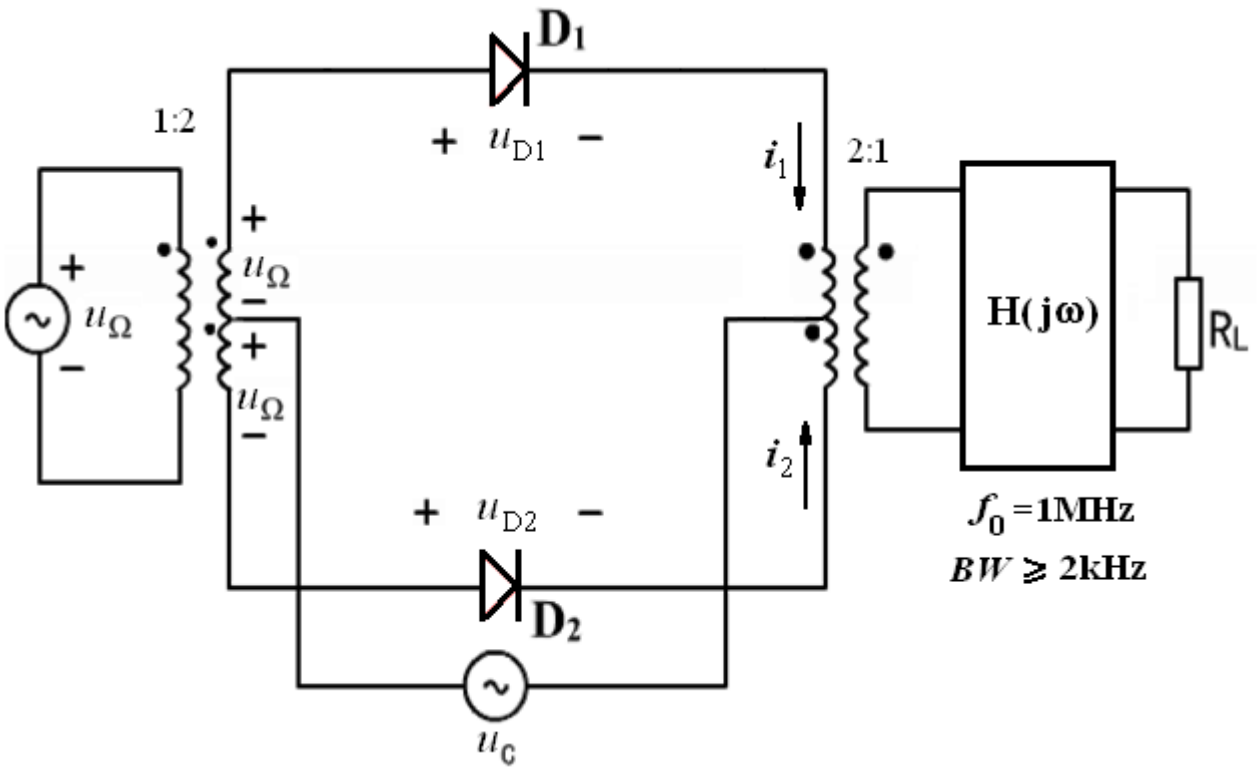
例： 二极管平衡调制器如图所示，已知

$$u_{\Omega} = U_{\Omega m} \cos 2\pi \times 10^{3} t \ \ (\mathrm{V})$$

$$u_{c} = U_{cm} \cos 2\pi \times 10^{6} t \ \ (\mathrm{V})$$

$$U_{cm} >> U_{\Omega m} \quad U_{cm} > 0.5 \ \ (\mathrm{V})$$

- 绘出 $u_o(t) = ?$
- 若 u_c 与 u_{Ω} 对调位置 $u_o(t) = ?$
- 若将此电路产生SSB信号，电路应如何变化?


D1
1:2
+ uD1 −
i1
2:1
+
uΩ
−
+
uΩ
−
H(jω)
RL
i2
+ uD2 −
f0 =1MHz
BW ≥ 2kHz
D2
uC

解

$$i = i_1 - i_2 = 2g_D U_{\Omega m}\cos\Omega t\left(\frac{1}{2} + \frac{2}{\pi}\cos\omega_c t - \frac{2}{3\pi}\cos 3\omega_c t + \cdots\right)$$

$$u_0(t) = \frac{4}{\pi} g_D U_{\Omega m} R_L \cos\Omega t \cos\omega_c t$$

经滤波后，输出电压 $u_o(t)$ 为

$$u_o(t) = i' R_L$$

$$= \frac{4}{\pi} g_D U_{\Omega m} R_L \cos(2\pi \times 10^3 t)\cos(2\pi \times 10^6 t) \ \ (\mathrm{V})$$

它是DSB信号

(2) $u_{D1} = u_\Omega + u_c \qquad u_{D2} = -(-u_\Omega + u_c) = -u_c + u_\Omega$

$$i_{D1} = g_D u_{D1} S(t) \quad i_{D2} = g_D u_{D2} S(t + \frac{T_c}{2})$$

$$i' = i_{D1} - i_{D2}$$

$$= g_D u_c \left[S(t) + S(t + \frac{T_c}{2}) \right] + g_D u_\Omega \left[S(t) - S(t + \frac{T_c}{2}) \right]$$

$$= g_D U_{cm} \cos \omega_c t + g_D U_{\Omega m} \cos \Omega t \left[\frac{4}{\pi} \cos \omega_c t - \frac{4}{3\pi} \cos 3\omega_c t + \cdots \right]$$

经滤波：

$$u_o(t) = i'R_L$$

$$= g_D R_L U_{cm} \cos \omega_c t + g_D R_L \frac{4}{\pi} U_{\Omega m} \cos \Omega t \cos \omega_c t$$

$$= U_{om}(1 + m_a \cos \Omega t) \cos \omega_c t$$

其中 $U_{om} = g_D R_L U_{cm}$ $\quad m_a = \dfrac{4U_{\Omega m}}{\pi U_{cm}}$

AM 信号

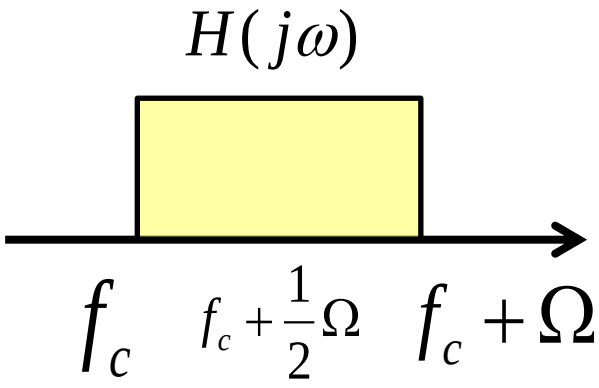
(3) 通过滤波器产生SSB信号。

上边频：$f_c + \Omega$

下边频：f_c

中心频率：$f_c + \frac{1}{2}\Omega$

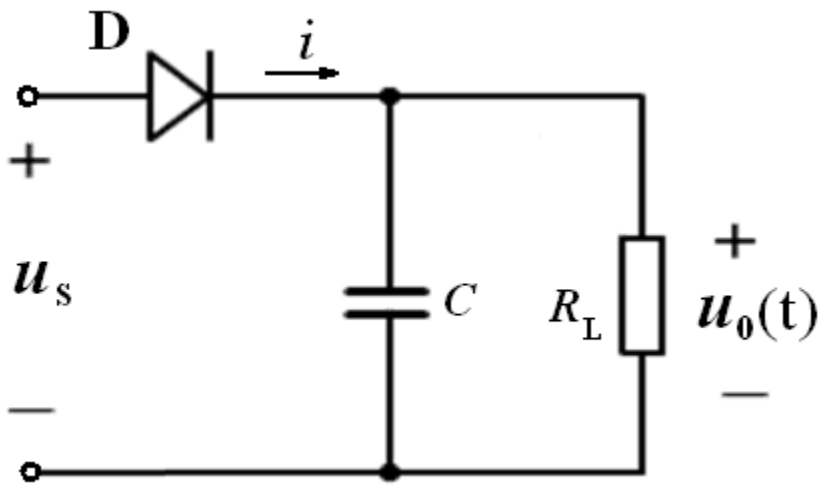
带宽：Ω



一. 大信号二极管包络检波

1. 工作原理： u_s 为AM波

$$u_s = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t \qquad \text{且} U_{im} > 0.5 \ (\text{V})$$

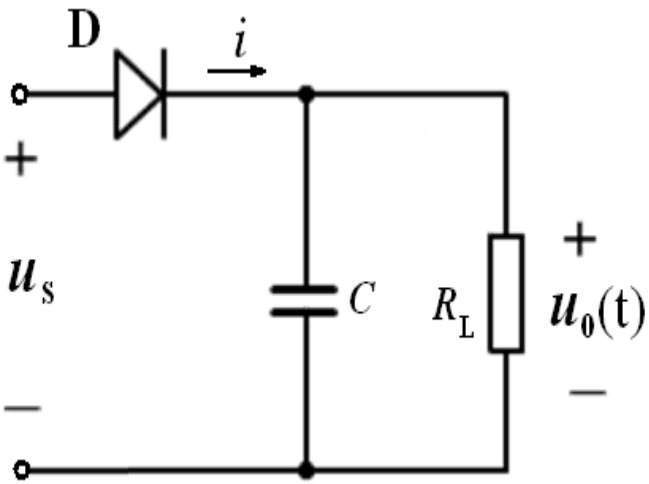


➢二极管导通：

充电时间常数 $R_D C \quad (R_D << R_L)$

充电时间常数越小，充电越快。

充电很快，很短时间就冲到峰值

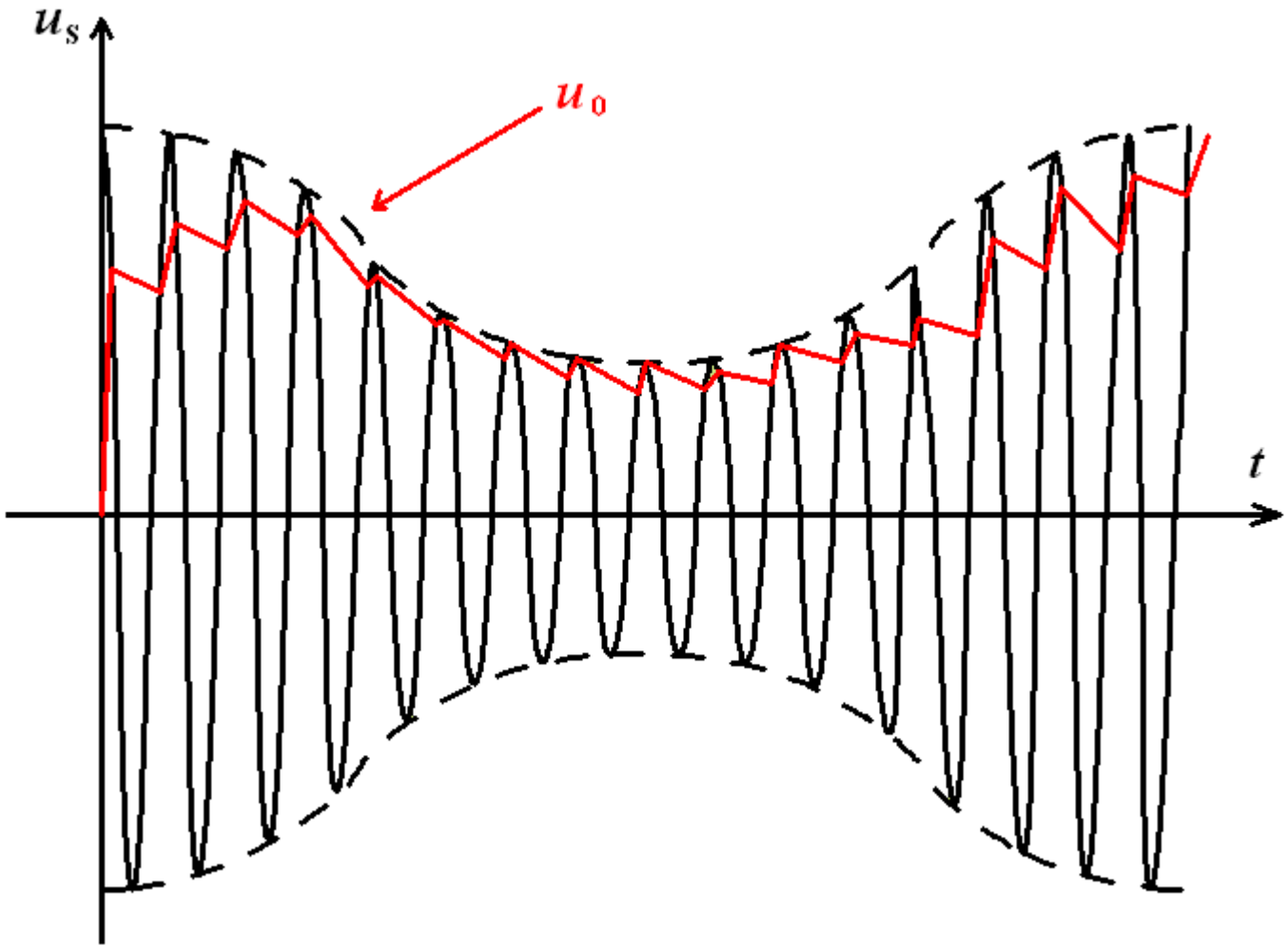


➢二极管截止：

放电时间常数 $R_L C$

放电时间常数越大，放电越慢。

放电较慢，时间较长


u_S
u_0
t

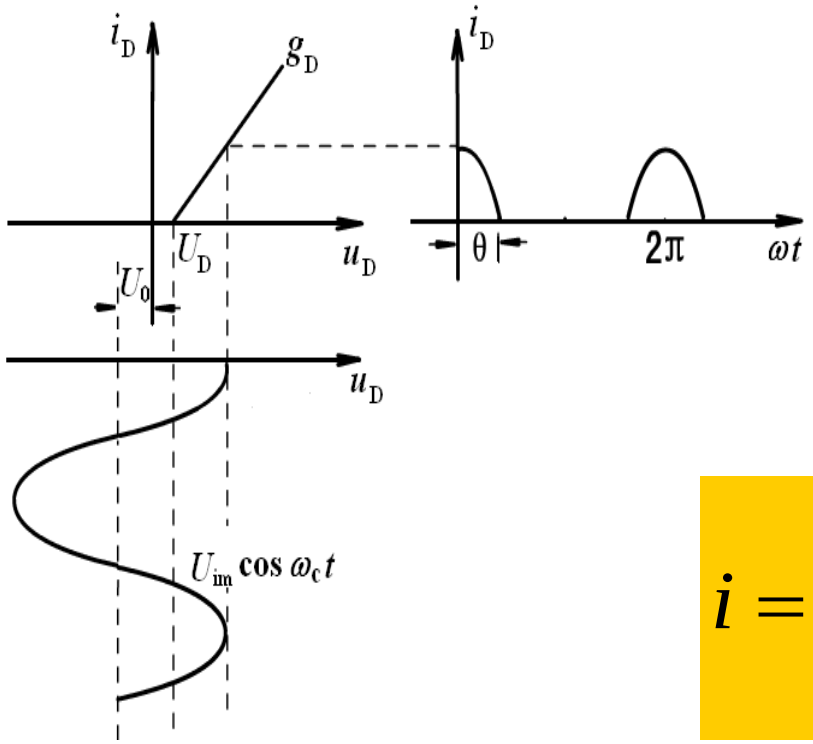
❖对Ω信号，C 相当开路

$$\frac{1}{\Omega C} >> R_L \rightarrow R_L C << \frac{1}{\Omega}$$

❖对ω_c信号，C 相当短路

$$\frac{1}{\omega_c C} << R_L \rightarrow R_L C >> \frac{1}{\omega_c}$$

由于 U_{im} 足够大，二极管的伏安特性折线近似。



$$i=\begin{cases} g_D(u_D-U_D) & u_D>U_D \\ 0 & u_D\le U_D \end{cases}$$

1） 若输入为载波，输出为直流

$$u_D = u_s - U_o = U_{im}\cos\omega_c t - U_o$$

$$i = g_D(u_D - U_D)$$

$$= g_D(U_{im}\cos\omega_c t - U_o - U_D)$$

当$\omega t = \theta$ $\quad i = 0$

$$\cos\theta = \frac{U_o + U_D}{U_{im}} \approx \frac{U_o}{U_{im}} \text{ 或 } U_o = U_{im}\cos\theta \quad ①$$

式(1)说明：输出的直流电压与导通角有关。

2）θ角仅由电路决定，与外加信号无关

（参考方程式1）

当$\omega t = 0$时，

$$I_m = g_D(U_{im} - U_o - U_D)$$
$$= g_D U_{im}(1 - \cos\theta)$$

直流量 $I_o = \alpha_o I_m = \dfrac{I_m(\sin\theta - \theta\cos\theta)}{\pi(1-\cos\theta)}$

$$= \frac{g_D U_{im}}{\pi}(\sin\theta - \theta\cos\theta) \qquad ②$$

输出平均电压 $U_o = R_L I_o$ ③

式 ① ② 代入式 ③

$$U_{im}\cos\theta = R_L \cdot \frac{g_D U_{im}}{\pi}(\sin\theta - \theta\cos\theta)$$

$$\frac{tg\theta - \theta}{\pi} = \frac{1}{g_D R_L}$$

$g_D R_L$ 很大，近似： $tg\theta \approx \theta + \frac{\theta^3}{3}$

得 $$\theta \approx \sqrt[3]{\frac{3\pi}{g_D R_L}}$$

2. 性能指标：

❖检波效率： $$K_d = \frac{U_o}{U_{im}} \approx \cos\theta$$

❖输入阻抗： $$\frac{1}{2} \cdot \frac{U_{im}^2}{R_i} = \frac{U_o^2}{R_L}$$ (载波功率近似等于输出功率)

$$\therefore \ R_i = \frac{1}{2} R_L$$

若输入为调幅波(AM)

$$u_s = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t$$

则输出 u_o 正比于 $U_{im}(1+m_a \cos\Omega t)$

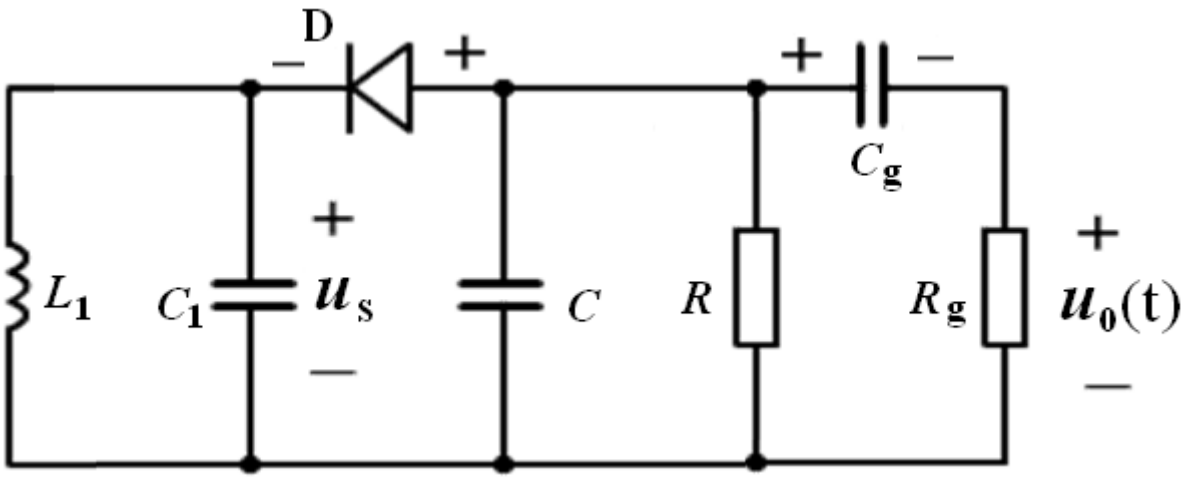
$$u_0 = K_d U_{im}(1 + m_a \cos \Omega t)$$

5-9.EXE

例1　　检波电路如图所示，

$$u_s = U_{sm}(1 + m\cos\Omega t)\cos\omega_c t \ ,\quad U_{sm} > 0.5\ \text{V}$$

画出RC两端，R_g两端，C_g 两端，二极管两端的电压波形。



解 $u_D = u_c - u_s$ 二极管 u_s 在负半周导

通，因此检波应是调幅波 u_s 的下包络。

$$u_c = K_d \times (u_s\text{下包络})$$

$$= K_d U_{sm}(1 + m_a \cos \Omega t)$$

$$u_D = u_c - u_s$$

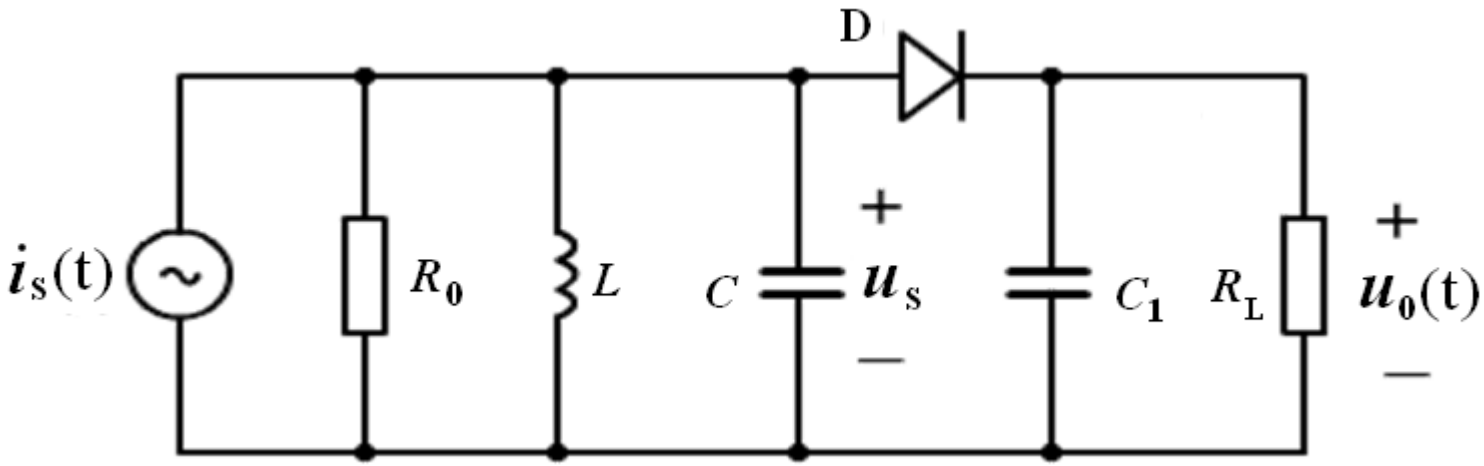
图解

例2 在题图所示的包络检波电路中，输入回路为并联谐振电路，其谐振频率 $f_o = 10^6$ Hz 谐振电阻 $R_o = 20$ kΩ，检波负载为10 kΩ，$R_D = 100$ Ω，$C_1 = 0.01$ μF

(1) 若 $i_s = 0.5\cos 2\pi \times 10^6 t$ (mA) 求检波器输入电压 u_s 及检波器输出电压 $u_o(t)$ 的表达式。

(2) 若 $i_s = 0.5(1 + 0.5\cos 2\pi \times 10^3 t)\cos 2\pi \times 10^6 t$ (mA) 求 $u_o(t)$ 的表达式

(1) 若 $i_s = 0.5\cos 2\pi\times10^6 t$ (mA) 求检波器输入电压 u_s 及检波器输出电压 $u_o(t)$ 的表达式。



(2) 若 $i_s = 0.5(1+0.5\cos 2\pi\times10^3 t)\cos 2\pi\times10^6 t$ (mA) 求 $u_o(t)$ 的表达式

解

（1） $\theta = \sqrt[3]{\dfrac{3\pi R_D}{R_L}} = \sqrt[3]{\dfrac{3\pi \times 100}{10 \times 10^3}} = 0.455\,\text{rad} = 26^\circ$

$$K_d = \cos\theta = \mathbf{0.9} \qquad R_i = \frac{R_L}{2} = 5\ \text{k}\Omega$$

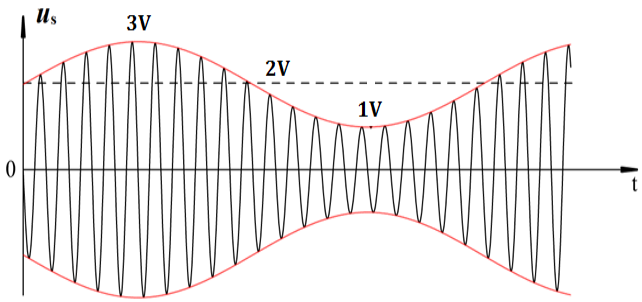
$$u_s = (R_o // R_i) \cdot i_s = (20 // 5) \cdot i_s$$

$$= 4 \times 0.5 \cos(2\pi \times 10^6)t = 2\cos(2\pi \times 10^6)t\ \ (\text{V})$$

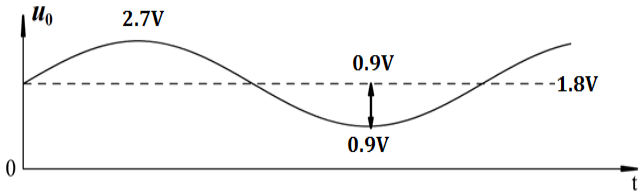
$$u_o(t) = K_d \cdot u_{sm}(t) = 1.8\ \text{V}$$

（2）

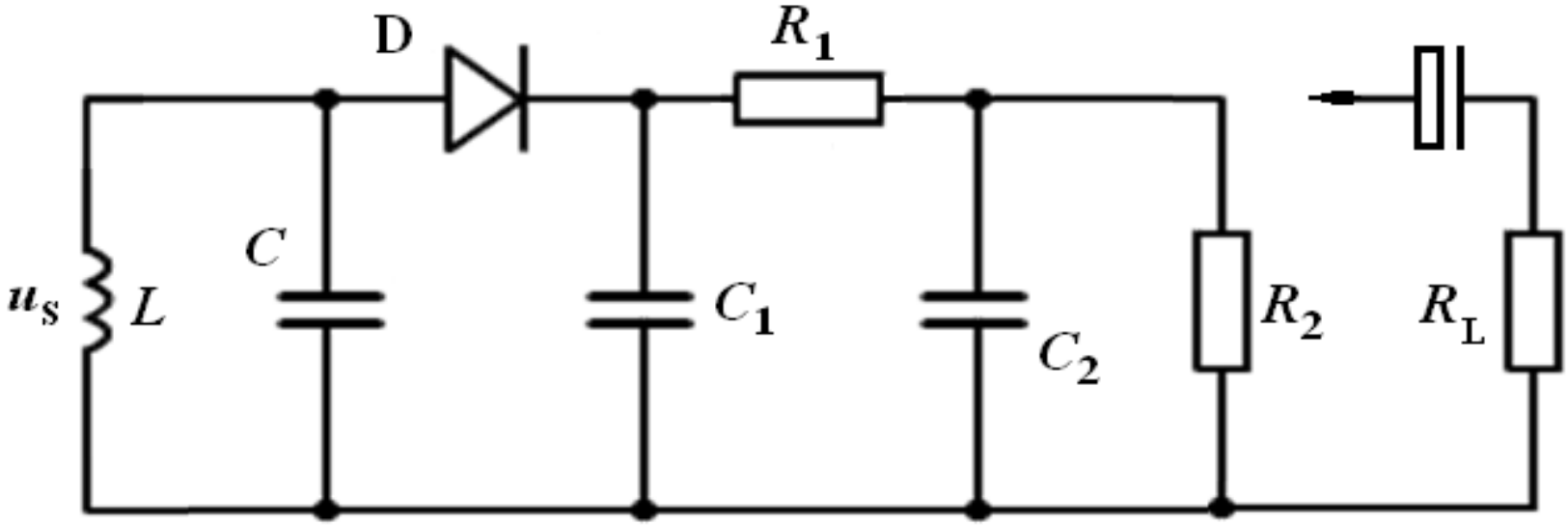
$$u_s = 4\times 0.5\left[1+0.5\cos(2\pi\times 10^3)t\right]\cos(2\pi\times 10^6)t$$
$$= 2\left[1+0.5\cos(2\pi\times 10^3)t\right]\cos(2\pi\times 10^6)t \quad \text{(V)}$$



$$u_o(t) = K_d U_{sm}(t)$$
$$= 1.8\left[1+0.5\cos(2\pi\times 10^3)t\right] \quad \text{(V)}$$



例4 如图二极管大信号峰值包络检波电路，已知输入电路的空载Q值：$Q_0 = 100$， 载波频率$f_0 = 465\ \text{kHz}$ $L = 560\ \mu\text{H}$ $R_L = 5\ \text{k}\Omega$ **若要求输入回路的通频带** $BW = 30\ \text{kHz}$ **不产生惰性失真，并且再接上交流负载后不产生负峰切割失真，试确定元件** R_1，R_2，C_1，C_2 **的值。**



$$u_s = U_{sm}(1+0.5\cos 2\pi\times 10^3 t)\cos 2\pi f_0 t$$

解

$$R_p = Q_0\omega_0 L = 100\times 2\pi\times 465\times 10^3\times 560\times 10^{-6}$$

$$= 163.53 \quad k\Omega$$

$$\frac{R_1+R_2}{2} // R_p = Q_L\omega_0 L = \frac{f_0}{\mathrm{BW}}\cdot\omega_0 L$$

$$= \frac{465\times 10^3}{30\times 10^3}\times 2\pi\times 465\times 10^3\times 560\times 10^{-6}$$

$$= 25 \ \mathrm{k\Omega}$$

设 $\dfrac{R_1+R_2}{2} = R_i \quad \Rightarrow \quad R_i \approx 29 \ \mathrm{k\Omega}$

$$\therefore \quad R_1 + R_2 = 2R_i = 58 \ \mathrm{k\Omega}$$

不产生惰性失真:

$$(R_1+R_2)(C_1+C_2)\le\frac{\sqrt{1-m_a^2}}{\Omega\cdot m_a}=\frac{\sqrt{1-0.5^2}}{2\pi\times10^3\times0.5}$$

$$\therefore\quad C_1+C_2\le4.755\ \mu\mathrm{F}$$

取 $C_1+C_2=4\ \mu\mathrm{F}$

$$C_1=C_2=\frac{C_1+C_2}{2}=2\ \mu\mathrm{F}$$

不产生负峰切割失真：

$$\frac{R_2 // R_g + R_1}{R_2 + R_1} \geq m_a = \mathbf{0.5}$$

$$\Rightarrow \begin{array}{l} R_1 \geq 24.66\ \text{k}\Omega \\ R_2 \leq 33.34\ \text{k}\Omega \end{array}$$

角度的调制和解调

一. 调频波、调相波：

1.调频波

$$\omega(t) = \omega_C + K_f u_\Omega(t)$$

瞬时角频率

未调制时载波的角频率

若设调制信号 $u_\Omega(t) = U_{\Omega m}\cos\Omega t$

$$\omega(t) = \omega_C + \boxed{K_f U_{\Omega m}}\cos\Omega t$$

$\Delta\omega_m$ → ω_C的偏移量

(最大角频率偏移与$U_{\Omega m}$成正比，与Ω无关)

$$\varphi(t) = \int_0^t \omega(t)dt + \varphi_0$$

$$= \omega_C t + \frac{K_f U_{\Omega m}}{\Omega}\sin\Omega t$$

令： $\varphi_0 = 0$

瞬时相位

m_f → **最大相位偏移，与Ω成反比**
调频指数

表达式： $u(t) = U_m\cos(\omega_C t + m_f \sin\Omega t)$

6-1.exe

2. 调相波

$$u(t) = U_m\cos(\omega_C t + m_P\cos\Omega t)$$

$$\varphi(t) = \omega_C t + K_P u_\Omega(t)$$

设调制信号：$u_\Omega(t) = U_{\Omega m}\cos\Omega t$

6-2.exe

$$\varphi(t) = \omega_C t + \underbrace{\boxed{K_P U_{\Omega m}}}_{m_P}\cos\Omega t$$

m_P 最大相位偏移，与$U_{\Omega m}$成正比、与Ω无关

调相指数

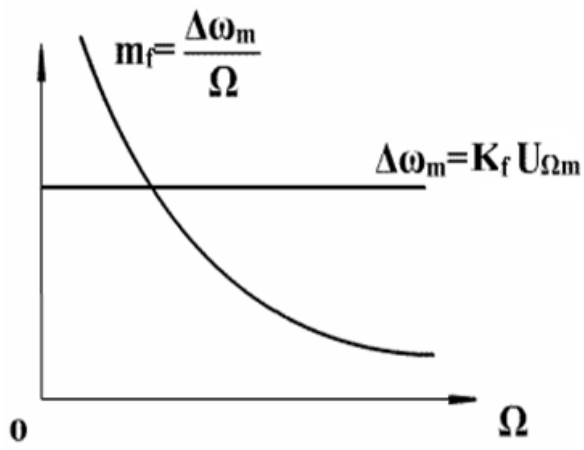
$$\omega(t) = \omega_C - \underbrace{\boxed{\Omega m_P}}_{\Delta\omega_m}\sin\Omega t$$

$\Delta\omega_m$ 与Ω成正比

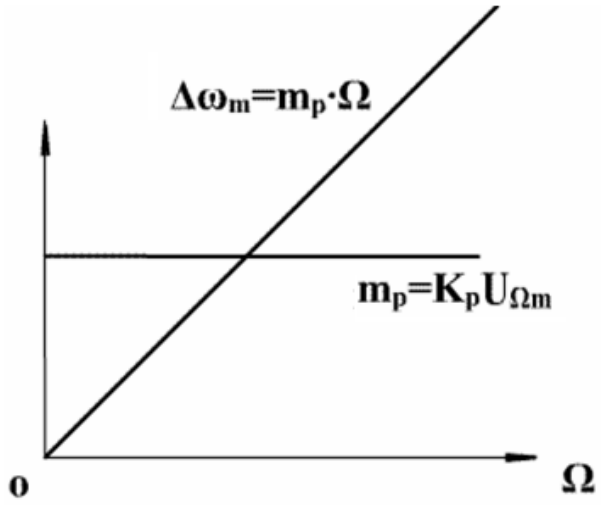
设调制信号 $u_\Omega(t) = U_{\Omega m}\cos\Omega t$ 情况下：

$$u(t) = U_m\cos(\omega_C t + m_f\sin\Omega t)$$ —— FM

$$u(t) = U_m\cos(\omega_C t + m_P\cos\Omega t)$$ —— PM



当 $U_{\Omega m}$ 恒定时，调频波



当 $U_{\Omega m}$ 恒定时，调相波

例1.

设有一组正弦调制信号，信号频率最低为 $F_{\min} = 300\text{Hz}$ 最高为 $F_{\max} = 3400\text{Hz}$。调制信号为等幅振荡。调频时，最大频偏 $\Delta f_m = 75\text{kHz}$；调相时，最大相偏 $m_p = 22\text{rad}$ 试求调频时调制指数 m_f 的变化范围及调相时最大频偏 Δf_m 的范围。

解

(1) 调频时： $U_{\Omega m}$不变，所以$\Delta\omega_m$不变，

而调频指数$m_f = \dfrac{\Delta\omega_m}{\Omega} = \dfrac{\Delta f_m}{F}$

$$m_{f\max} = \frac{\Delta f_m}{F_{\min}} = \frac{75\times10^3}{300} = 250$$

$$m_{f\min} = \frac{\Delta f_m}{F_{\max}} = \frac{75\times10^3}{3400} \approx 22$$

$\therefore$ 调频时，$\Delta\omega_m = k_f U_{\Omega m}$不变，

但$m_f \propto \dfrac{1}{\Omega}$,有很大变化。

(2) 调相时： $U_{\Omega m}$不变，所以m_p不变，

$$\Delta f_m = m_p F$$

$$\Delta f_{m\max} = m_p F_{\max} = 22 \times 3400 = 74.8\text{kHz}$$

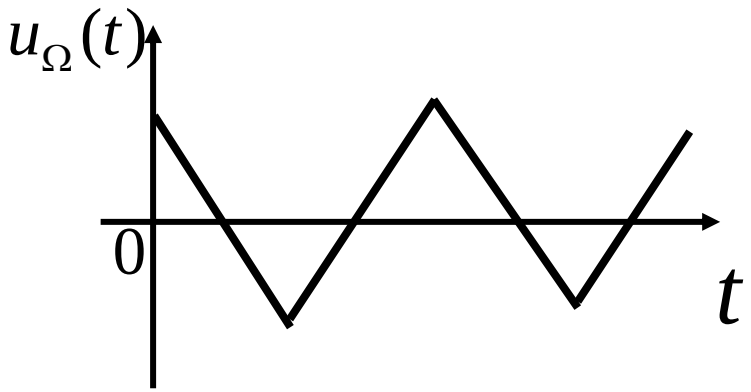
$$\Delta f_{m\min} = m_p F_{\min} = 22 \times 300 = 6.6\text{kHz}$$

$\therefore$ 调相时，$m_p = k_p U_{\Omega m}$不变

但$\Delta f_m \propto F$有很大变化

例2.

调制信号$u_\Omega(t)$ 为周期重复的三角波。画出调频波和调相波.



6-3.exe

6-4.exe

二. 调角信号的频谱

FM、PM波的数学表示式是相似的，统一写成调角波表示式

$$u(t) = U_m \cos(\omega_c t + m \sin \Omega t)$$

利用三角函数展开：

$$u(t) = U_m [\cos(m \sin \Omega t) \cos \omega_c t - \sin(m \sin \Omega t) \sin \omega_c t]$$

在贝塞尔函数理论中，已经证明存在下列关系：

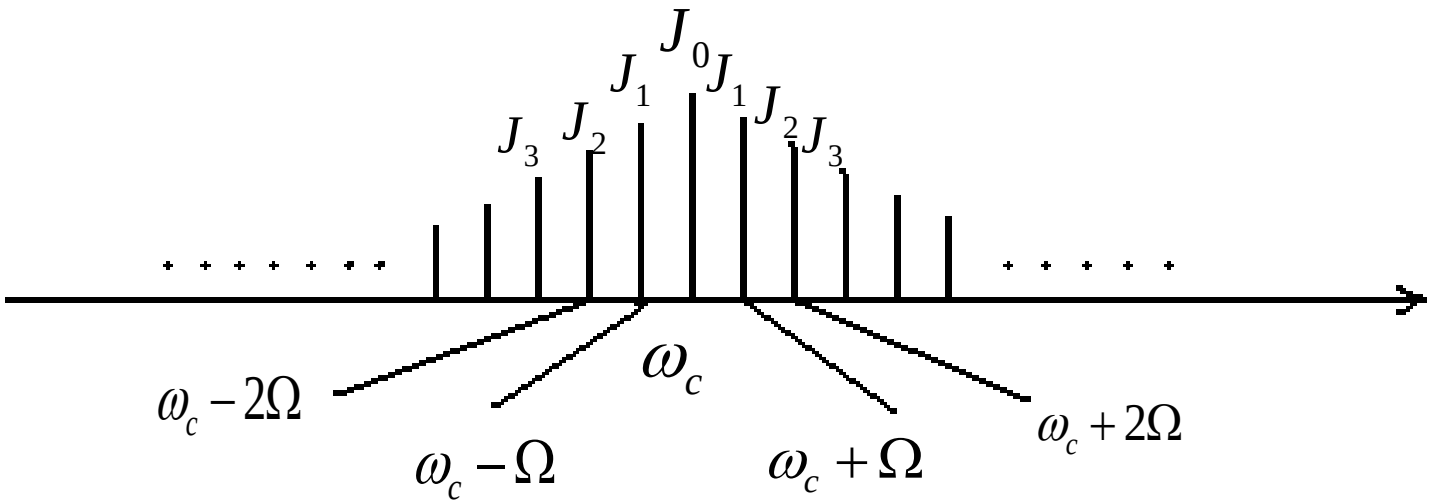
$$\cos(m \sin \Omega t) = J_0(m) + 2J_2(m) \cos 2\Omega t + 2J_4(m) \cos 4\Omega t + \ldots\ldots$$

$$\sin(m\sin\Omega t) = 2J_1(m)\sin\Omega t + 2J_3(m)\sin 3\Omega t$$
$$+ 2J_5(m)\sin 5\Omega t + \ldots\ldots$$

式中$J_n(m)$是以m为宗数的n阶第一 类贝塞尔函数。代入上式，整理得：

$$u(t) = U_m J_0(m)\cos\omega_c t + U_m J_1(m)[\cos(\omega_c + \Omega)t - \cos(\omega_c - \Omega)t]$$
$$+ U_m J_2(m)[\cos(\omega_c + 2\Omega)t + \cos(\omega_c - 2\Omega)t]$$
$$+ U_m J_3(m)[\cos(\omega_c + 3\Omega)t - \cos(\omega_c - 3\Omega)t] + \ldots\ldots$$

调角的频谱为：



1) $\omega_c \pm n\Omega$无限多对上、下边频分量 $U_m J_n(m)$

当n为偶数时，上、下边频分量相加

当n为奇数时，上、下边频分量相减

第一类贝塞尔函数

调制指数	载波	边频对												
m	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}	J_{11}	J_{12}	J_{13}
0.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-
0.25	0.98	0.12	-	-	-	-	-	-	-	-	-	-	-	-
0.5	0.94	0.24	0.03	-	-	-	-	-	-	-	-	-	-	-
1.0	0.77	0.44	0.11	0.02	-	-	-	-	-	-	-	-	-	-
1.5	0.51	0.56	0.23	0.06	0.01	-	-	-	-	-	-	-	-	-
2.0	0.22	0.58	0.35	0.13	0.03	-	-	-	-	-	-	-	-	-
2.4	0	0.52	0.43	0.20	0.06	0.02	-	-	-	-	-	-	-	-
2.5	-0.05	0.50	0.45	0.22	0.07	0.02	0.01	-	-	-	-	-	-	-
3.0	-0.26	0.34	0.9	0.31	0.13	0.04	0.01	-	-	-	-	-	-	-
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02	-	-	-	-	-	-
5.0	-0.18	-0.33	0.5	0.36	0.39	0.26	0.13	0.05	0.02	-	-	-	-	-
5.52	0	-0.34	-0.12	0.26	0.40	0.32	0.19	0.09	0.03	0.01	-	-	-	-
6.0	0.15	-0.28	-0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02	-	-	-	-
7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02	-	-	-
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03	-	-
8.65	0	0.27	0.06	-0.24	-0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02	-
9.0	-0.09	0.25	0.14	-0.18	-0.27	-0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01
10.0	-0.25	0.05	0.25	0.06	-0.22	-0.23	-0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03

2) 当调制指数 m 增大时，则具有较大振幅的边频分量也增加。

3) 当m为某些特定值时，如m=2.405、5.520、8.653、……载波分量成分 $J_0(m)=0$。用来测量频偏 $\Delta\omega_m$ 和调制指数 m_f 。

三、调角信号的频谱宽度

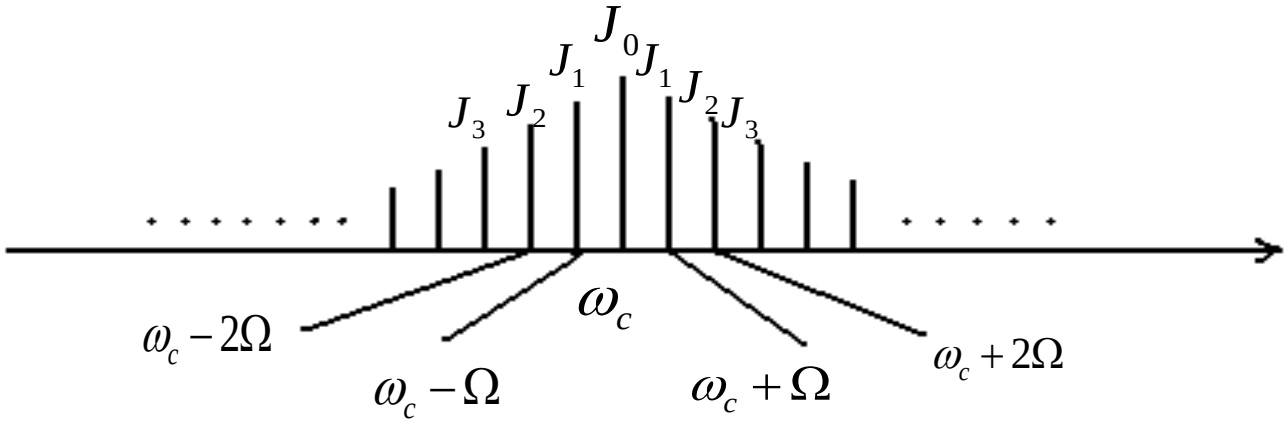
1) $m<1$,窄带调制 m很小，近似：$\cos(m\sin\Omega t)\approx 1$

$$\sin(m\sin\Omega t)\approx m\sin\Omega t$$

$$u(t)=U_m\cos(\omega_c t)+\frac{m}{2}U_m\cos[(\omega_c+\Omega)t]-\frac{m}{2}U_m\cos[(\omega_c-\Omega)t]$$

调角信号的频谱与AM波频谱相似。由ω_c、$\omega_c\pm\Omega$构成。差别就是下边频的相位相反。频谱宽度为 2Ω rad/s. 它广泛运用于移动通信中。

2) 宽带调制 $m>1$



频谱是无限宽的，认为当边频幅度小于载频的10%时，即忽略这些边频。因此，实际调角信号所占的有效频谱宽度是有限的。当$m>1$时，$m+1$ 以上各阶频谱的幅度均小于载波幅度10%。因而可以忽略。

有效频谱宽度为：

$$BW = 2(m+1)F = 2(\Delta f_m + F)\text{Hz} \qquad F = \frac{\Omega}{2\pi}$$

以上讨论的是单音调制情况。

实际上，调制信号中包括很多频率分量，设最高频率 $F_{\max}$

$$BW = 2(m+1)F_{\max} = 2(\Delta f_m + F_{\max})\text{Hz}$$

FM与PM的比较：

$$(BW)_{\text{FM、PM}} = 2(m+1)\Omega$$

但当Ω增加时，调频:

$m_f = \frac{K_f U_{\Omega m}}{\Omega}$减小，$BW_{\text{FM}} = 2(m_f + 1)\Omega$略有增加；

调相:

$m_p = K_p U_{\Omega m}$不变，

$BW_{\text{PM}} = 2(m_p + 1)\Omega$ 与Ω正比例增加.

3. 功率

AM: $$P_{平均} = (1 + \frac{1}{2} m_a{}^2) P_{载波}$$

$$m_a = 1: \quad P_{平均} = (1 + \frac{1}{2}) P_{载波} = 1.5 P_{载波}$$

FM、PM:

调角波振幅不变，工作在最大工作 状态.

功率: $$P_f = \frac{U_m{}^2}{2} \sum_{n=-\infty}^{\infty} J_n{}^2(m)$$

根据第一类贝塞尔函数特性：

$$\therefore P_f = \frac{U_m^{\ 2}}{2}$$

$$\sum_{n=-\infty}^{\infty} J_n^{\ 2}(m) = 1$$

上式表明，当Um一定时，不论 m_f 为何值，调频波的平均功率恒为定值，并且等于未调制的载波频率。换句话说，m_f 改变仅会引起载波分量和各边带分量之间功率的重新分配，但不会引起总功率的改变。

例3. 已知 $u_\Omega = U_{\Omega m}\cos 2\pi \times 10^3 t, m_f = m_p = 10,$

求(1)此时FM波和PM波的带宽?

(2)若$U_{\Omega m}$不变, F 增大一倍两种调制信号的带宽如何?

(3)若F 不变, $U_{\Omega m}$增大一倍两种调制信号的带宽如何?

(4)若$U_{\Omega m}$、F都增大一倍两种调制信号的带宽如何?

解:(1) $\because u_{\Omega} = U_{\Omega m}\cos 2\pi \times 10^3 t, \therefore F = 10^3\,\text{Hz}$

$$\text{FM}: BW_{\text{FM}} = 2(m_f + 1)F = 2(10+1)\times 10^3 = 22\text{kHz}$$

$$\text{PM}: BW_{\text{PM}} = 2(m_p + 1)F = 2(10+1)\times 10^3 = 22\text{kHz}$$

(2)FM: $U_{\Omega m}$不变，则$\Delta f_m = K_f U_{\Omega m}$不变；$m_f = \dfrac{\Delta f_m}{F} = 5$

$$BW_{\text{FM}} = 2(m_f + 1)F = 2(5+1)\times 2\times 10^3 = 24\text{kHz}$$

PM: $U_{\Omega m}$不变，m_p不变

$$BW_{\text{PM}} = 2(10+1)\times 2\times 10^3 = 44\text{kHz}$$

(3)FM：F不变，$U_{\Omega m}$增大一倍，

则$\Delta f_m = K_f U_{\Omega m}$增大一倍；$m_f = 20$增大一倍。

$$BW_{\mathrm{FM}} = 2(20+1)\times 10^3 = 42\mathrm{kHz}$$

PM：m_p也增大一倍,$(m_p = K_p U_{\Omega m} = 20)$

$$BW_{\mathrm{PM}} = 2(20+1)\times 10^3 = 42\mathrm{kHz}$$

(4)$F, U_{\Omega m}$均增大一倍，FM：m_f不变,$(m_f = \dfrac{\Delta f_m}{F})$

$$BW_{\mathrm{FM}} = 2(m_f+1)F = 2(10+1)\times 2\times 10^3 = 44\mathrm{kHz}$$

PM：m_p增大一倍，$m_p = 20$

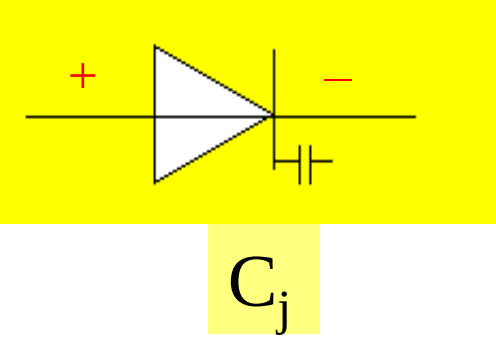
$$BW_{\mathrm{PM}} = 2(20+1)\times 2\times 10^3 = 84\mathrm{kHz}$$

二. 变容二极管直接调频电路：

利用它的结电容受反向外加电压控制而变化来实现调频。

1. 变容二极管

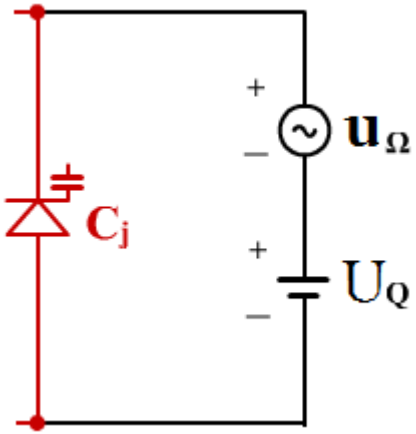
$$C_j = \frac{C_{j0}}{(1-\frac{u}{U_D})^{\gamma}}$$



C_{j0} —— 反向电压u = 0 时的结电容

U_D —— PN结内建电位差(很小)

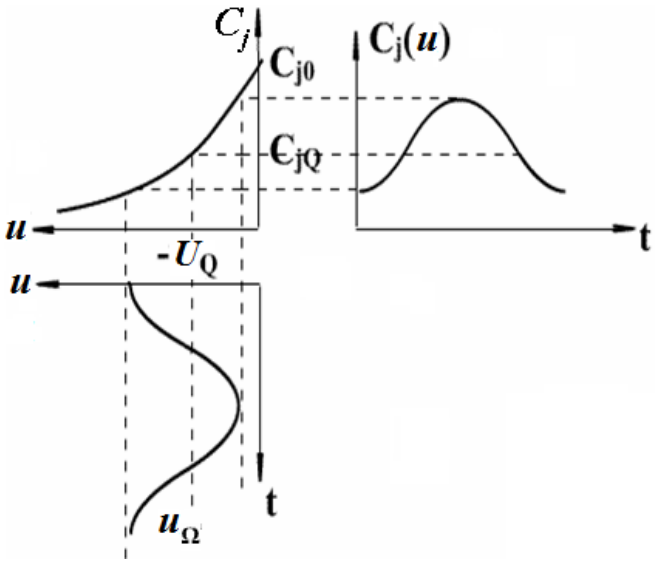
γ —— 电容的变化指数



当调制信号为：

$$u_\Omega(t) = U_{\Omega m}\cos\Omega t$$

$$C_j = \frac{C_{j0}}{\left(1+\dfrac{U_Q + U_{\Omega m}\cos\Omega t}{U_D}\right)^\gamma}$$



$$= \frac{C_{j0}}{\left(\dfrac{U_D + U_Q}{U_D}\right)^\gamma \left(1+\dfrac{U_{\Omega m}}{U_D + U_Q}\cos\Omega t\right)^\gamma}$$

→ C_{jQ}

静态时的结电容：

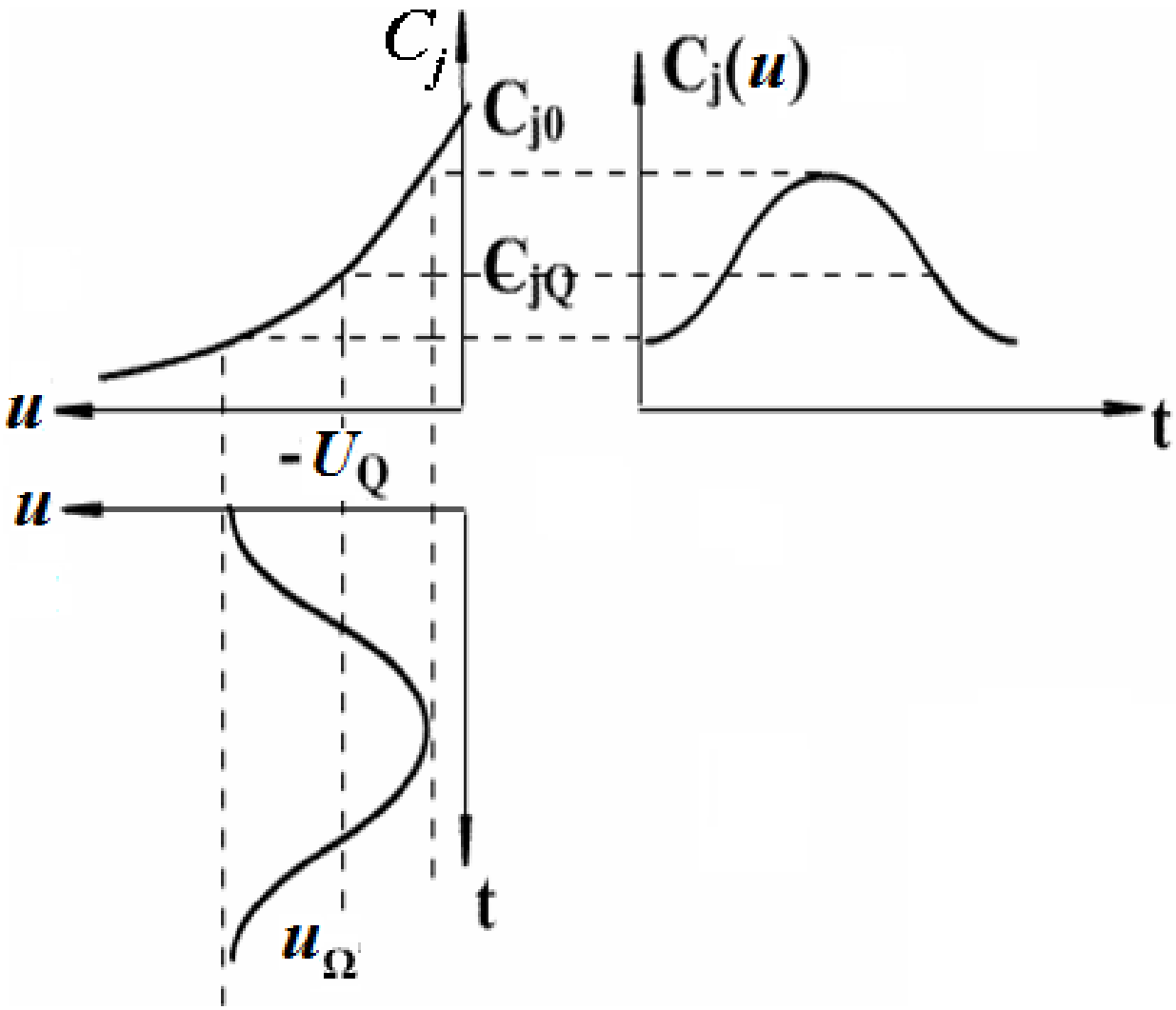
当$u=-U_Q$时，$C_{jQ}=\dfrac{C_{j0}}{(1+\dfrac{U_Q}{U_D})^{\gamma}}$

$$C_j=\frac{C_{j0}}{(\dfrac{U_D+U_Q}{U_D})^{\gamma}(1+\dfrac{U_{\Omega m}}{U_D+U_Q}\cos\Omega t)^{\gamma}}$$

$$C_j=\frac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}$$

$$M=\frac{U_{\Omega m}}{U_D+|U_Q|}$$

结电容的调制度


C_j
C_{j0}
$C_j(u)$
C_{jQ}
u
t
$-U_Q$
u
t
u_Ω

2. 变容管作振荡回路的总电容

6-5.exe

大频偏直接调制电路

C_1 ——— 隔直作用

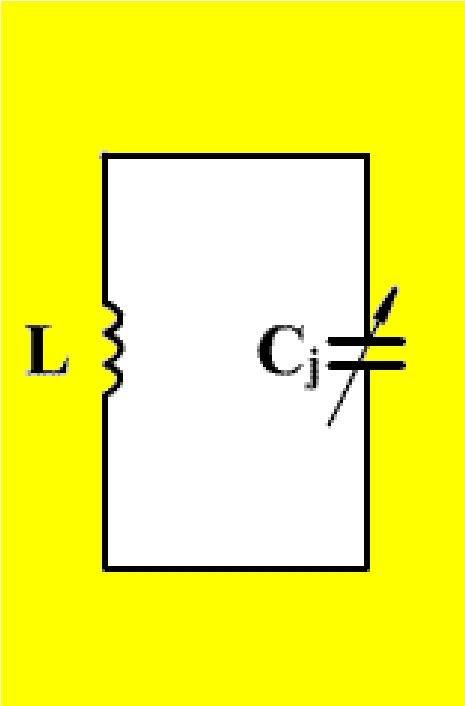
C_2 ——— 高频滤波电容

（对高频短路，对Ω开路）

L_1 ——— 高频扼流圈

应满足： $|U_Q| \geq U_{\Omega m}$

瞬时振荡角频率为:



$$\omega(t)=\frac{1}{\sqrt{LC_j}}=\frac{1}{\sqrt{L\dfrac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}}}$$

$$=\omega_c(1+M\cos\Omega t)^{\frac{\gamma}{2}}$$

上式称为调制特性方程

$\omega_c=\dfrac{1}{\sqrt{LC_{jQ}}}$ 是静态工作点($u_\Omega=0$)的振荡角频率。

调制特性分析:

(1) $\gamma = 2$

$$\omega(t) = \omega_c + M\omega_c \cos\Omega t$$

线性调制

这时，调频波获得最大角频偏为 $\Delta\omega_m = M\omega_c$

(2) $\gamma \neq 2$

$$\omega(t) = \omega_c[1 + \frac{\gamma}{2} M \cos\Omega t + \frac{\gamma}{2}(\frac{\gamma}{2} - 1)\frac{1}{2!} M^2 \cos^2\Omega t]$$

$$= \omega_c[1 + \frac{1}{8}\gamma(\frac{\gamma}{2} - 1)M^2 + \frac{\gamma}{2} M \cos\Omega t + \frac{1}{8}\gamma(\frac{\gamma}{2} - 1)M^2 \cos 2\Omega t]$$

中心频率偏移 **基波分量** **两次谐波分量**

最大角频偏 $\Delta\omega_m = \frac{\gamma}{2} M\omega_c$

最大角频偏

$$\Delta\omega_m = \frac{\gamma}{2} M\omega_c$$

$$k_f = \frac{\Delta f_m}{U_{\Omega m}} = \frac{\gamma}{2}\frac{Mf_c}{U_{\Omega m}}$$ 调制灵敏度

$$\Delta\omega_c \approx \frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2\omega_c$$ 中心角频率偏离ω_c的数值

$$\frac{\Delta\omega_c}{\omega_c} \approx \frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2$$ 中心角频率的相对偏离值

当γ一定，增大M,可增大频偏$\Delta\omega_m$,但同时也增大了$\Delta\omega_c$和$\frac{\Delta\omega_c}{\omega_c}$

若M一定,提高ω_c,可增大调频波最大偏值$\Delta\omega_m$

例 调频振荡回路有电感L和变容二极管组成。$L=2\mu\text{H}$,

变容二极管的参数为: $C_{j0}=225\text{pF}, U_D=0.6\text{V}, \gamma=\dfrac{1}{2}$

$U_Q=-6\text{V}$, 调制$u_\Omega(t)=3\sin(10^4 t)\text{V}$, 输出FM波。求:

(1)载波f_c

(2)由调制信号引起的载波漂移Δf_c

(3)最大频偏Δf_m

(4)调频系数（调制灵敏度）k_f

(5)二阶失真系数

解

$$C_j = \frac{C_{j0}}{(1+\frac{u}{U_D})^{\gamma}} = \frac{C_{j0}}{(1+\frac{u_\Omega + |u_Q|}{U_D})^{\gamma}}$$

$$= \frac{225}{(1+\frac{3\cos 10^4 t + 6}{0.6})^{\frac{1}{2}}}$$

$$= \frac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}$$

$$\because C_{jQ} = \frac{C_{j0}}{(1+\frac{U_Q}{U_D})^{\gamma}} = \frac{225}{(1+\frac{6}{0.6})^{\frac{1}{2}}} = 67.8\text{pf} \quad M = \frac{U_{\Omega m}}{U_D + U_Q} = 0.5$$

$$\therefore C_j = \frac{67.8}{(1+0.5\cos 10^4 t)^{\frac{1}{2}}}$$

$$\omega(t)=\omega_c(1+M\cos\Omega t)^{\frac{\gamma}{2}}\approx\omega_c+\Delta\omega_c+\Delta\omega_m\cos\Omega t$$

$$+\Delta\omega_{2m}\cos 2\Omega t$$

$$\omega_c=\frac{1}{\sqrt{LC_{jQ}}}=\frac{1}{\sqrt{2\times10^{-6}\times67.8\times10^{-12}}}\approx85.9\times10^6\,\text{rad/s}$$

$$\Delta\omega_m=\frac{\gamma}{2}M\omega_c=\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times85.9\times10^6\approx10.7\times10^6\,\text{rad/s}$$

$$\Delta\omega_{2m}=\Delta\omega_c=\frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2\omega_c$$

$$=\frac{1}{8}\times\frac{1}{2}\times(\frac{1}{4}-1)\times\frac{1}{4}\times85.9\times10^6\approx-10^6\,\text{rad/s}$$

$$(1)\ f_c=\frac{\omega_c}{2\pi}\approx13.7\text{MHz}$$

$$(2)\Delta f_c = \frac{\Delta\omega_c}{2\pi} = \frac{10^6}{2\pi} \approx 159\text{kHz}$$

$$(3)\Delta f_m = \frac{\Delta\omega_m}{2\pi} = \frac{10.7\times10^6}{2\pi} \approx 1.7\text{MHz}$$

$$(4)k_f = \frac{\Delta f_m}{U_{\Omega m}} = \frac{1.7\times10^6}{3} = 5.7\times10^5\,\text{Hz/V}$$

$$(5)k_{f2} = \frac{\Delta\omega_{2m}}{\Delta\omega_m} = \left|\frac{M}{4}(\frac{\gamma}{2}-1)\right| = \left|\frac{1}{8}(\frac{1}{4}-1)\right| = 0.094$$

3. 变容管部分接入振荡回路

6-6.exe

小频偏的直接调频电路

C_3为高频滤波电容

$$C = C_1 + \frac{C_2 C_j}{C_2 + C_j} = C_1 + \frac{C_2 C_{jQ}}{C_2 (1 + M\cos\Omega t)^{\gamma} + C_{jQ}}$$

$$\omega(t) = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L[C_1 + \frac{C_2 C_{jQ}}{C_2 (1 + M\cos\Omega t)^{\gamma} + C_{jQ}}}}$$

展开成幂级数，只考虑二次方项：

$$\omega(t)=\omega_c(1+A_1M\cos\Omega t+A_2M^2\cos^2\Omega t)$$

$$=\omega_c+\frac{A_2}{2}M^2\omega_c+A_1M\omega_c\cos\Omega t+\frac{A_2}{2}M^2\omega_c\cos 2\Omega t$$

$$\omega_c=\frac{1}{\sqrt{L(C_1+\frac{C_2C_{jQ}}{C_2+C_{jQ}})}}$$

$$A_1=\frac{\gamma}{2P}\qquad A_2=\frac{3\gamma^2}{8P}+\frac{1}{4}\frac{\gamma(\gamma-1)}{P}-\frac{1}{2}\frac{\gamma^2}{P(1+P_1)}$$

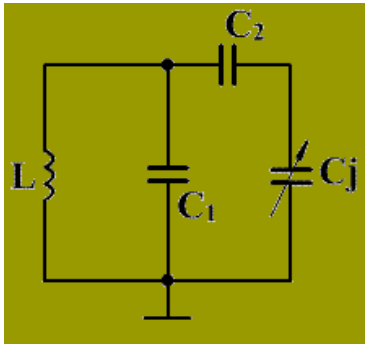
其中： $P=(1+P_1)(1+P_2+P_1P_2)$

$$p_1 = \frac{C_{jQ}}{C_2}$$

$$\frac{C_{jQ}}{C_{串}}$$

$$p_2 = \frac{C_1}{C_{jQ}}$$

$$\frac{C_{并}}{C_{jQ}}$$



最大频偏：$\Delta f_m = \frac{\gamma}{2P} M f_c$

中心频率偏移：$\Delta f_c = \frac{A_2}{2} M^2 f_c$

讨论：

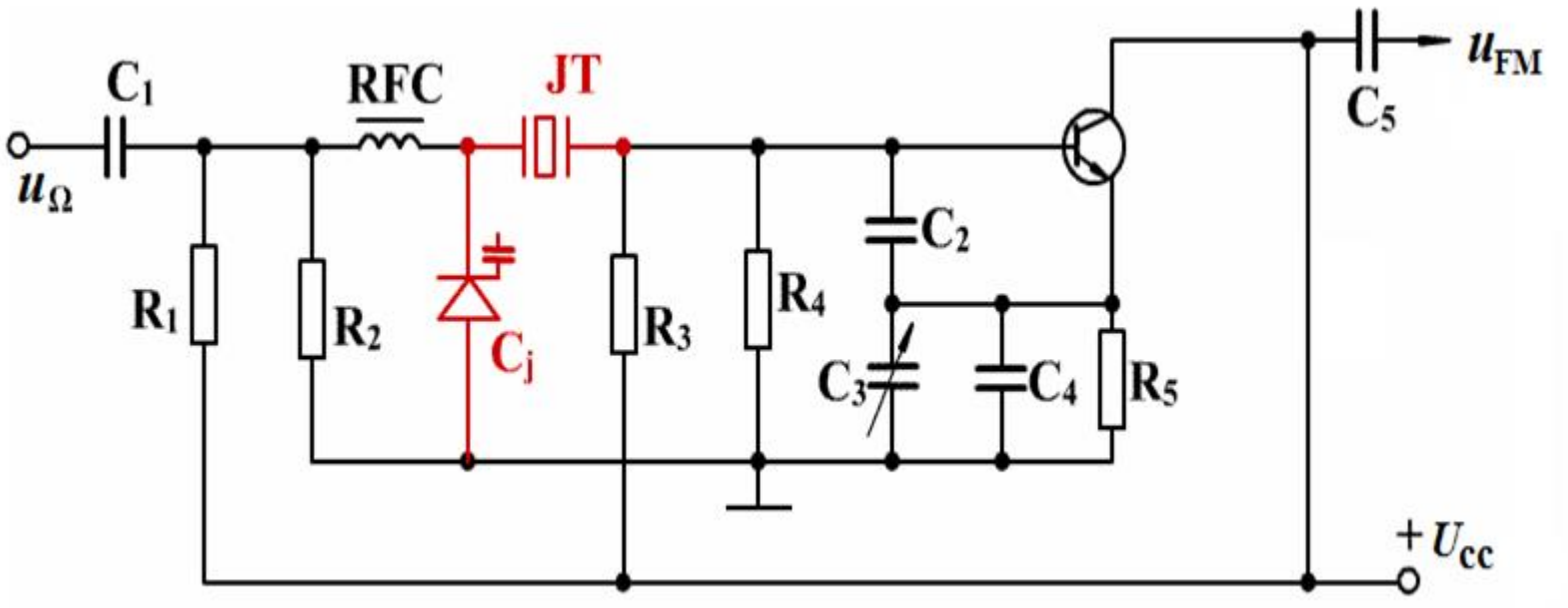
$$\left.\begin{array}{l} C_2\downarrow \quad P_1\uparrow \\ C_1\uparrow \quad P_2\uparrow \end{array}\right\} \longrightarrow P\uparrow \longrightarrow \Delta\omega_m\downarrow$$

C_1、C_2 的引入，使 C_j 对回路总电容的影响减小，从而 ω_c 的稳定性提高

例：题图所示电路采用2CC-401A变容管，已知变容管

$C_{j0}=109\text{pF}, U_D=1\text{V}, U_Q=-3.5\text{V}, \gamma=\frac{1}{2}$，石英晶体的$C_q=33.7\times10^{-3}\text{pF}$，音频调制到电压$U_{\Omega m}=1.4\text{V}$，振荡器的载频$f_c=12.5\text{MHz}$.

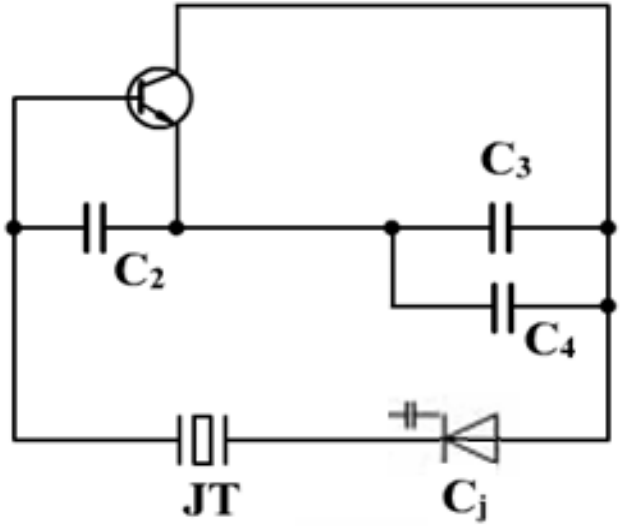
试指出各元器件的作用；画出高频等效电路；计算最大频偏Δf_m=?



各元器件的作用略。

解

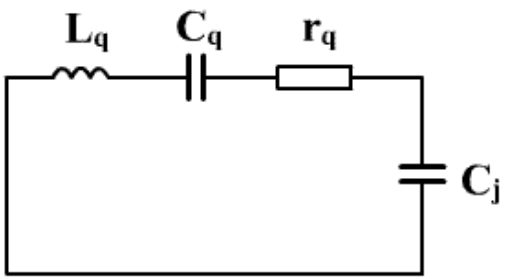
等效电路图



克拉泼振荡器

$$M=\frac{U_{\Omega m}}{U_D+|U_Q|}=\frac{1.4}{1+3.5}\approx 0.311$$

$$C_{jQ}=\frac{C_{j0}}{(1-\frac{U_Q}{U_D})^{\gamma}}=\frac{109}{(1+3.5)^{\frac{1}{2}}}\approx 51.4\text{pf}$$



$$P_2 = \frac{C_{并}}{C_{jQ}} = 0$$

由图可知：$P_1 = \dfrac{C_{jQ}}{C_{串}} = \dfrac{C_{jQ}}{C_q} = \dfrac{51.4}{33.7\times10^{-3}} \approx 1525$

则$P = 1 + P_1 = 1526$

最大频偏：

$$\Delta f_m = \frac{\gamma}{2P} M f_c = \frac{1/2}{2\times1526}\times0.311\times12.5\times10^6$$

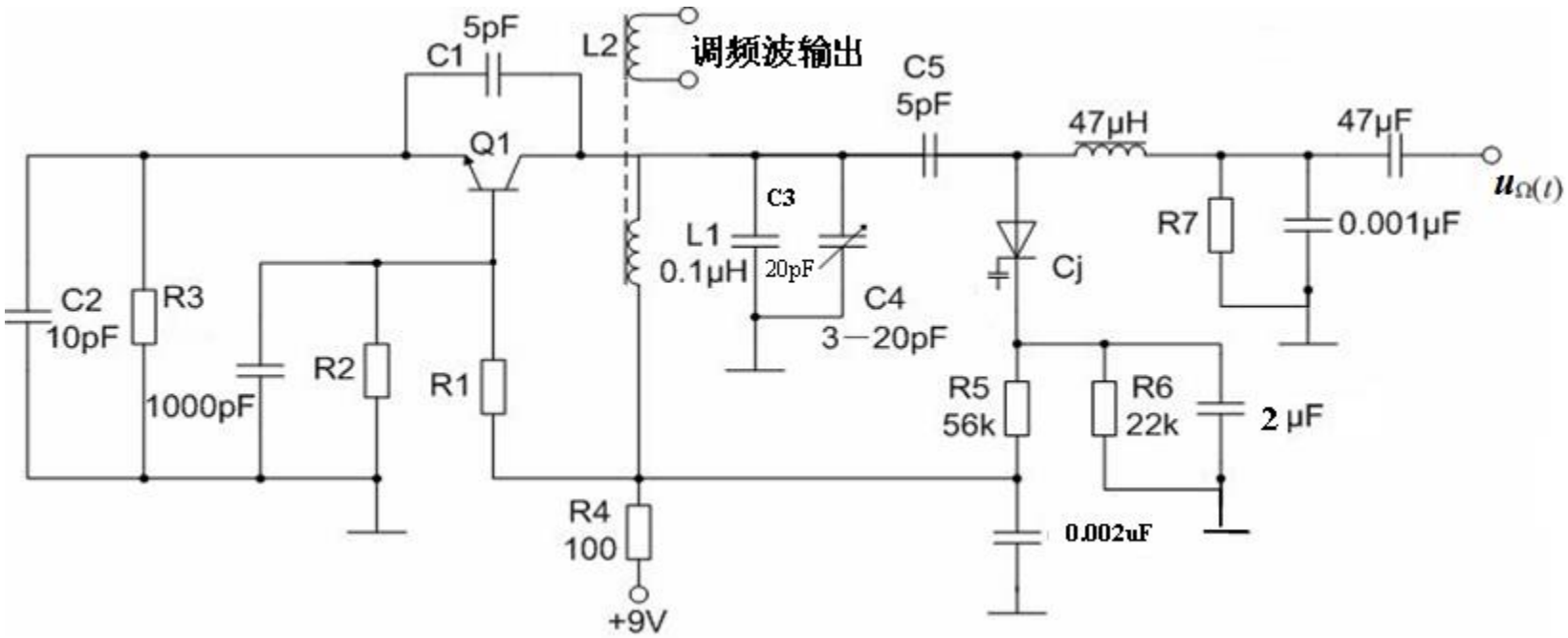
$$\approx 636.8\text{Hz}$$

例：对题图所示的变容管直接调频电路。

（1）说明电路中各元件的作用；

（2）画出该调频振荡器的高频通路、变容管的直流通路和音频通路。

（3）若已知 $u_\Omega = U_{\Omega m}\cos 2\pi\times 10^4 t(\mathrm{V})$，变容管结电容 $C_j = 80(U_Q + u_\Omega)^{-\frac{1}{2}}(\mathrm{pf})$，调频指数 $m_f = 5$，当 $u_\Omega = 0$ 时的振荡频率为 $f_c = 90\mathrm{MHz}$，试求变容管所需的直流偏置电压 U_Q，最大频偏 Δf_m 以及调制信号的振幅 $U_{\Omega m}$。



解

(1) R_1、R_2、R_3、R_4为晶体管提供直流偏置和直流通路;

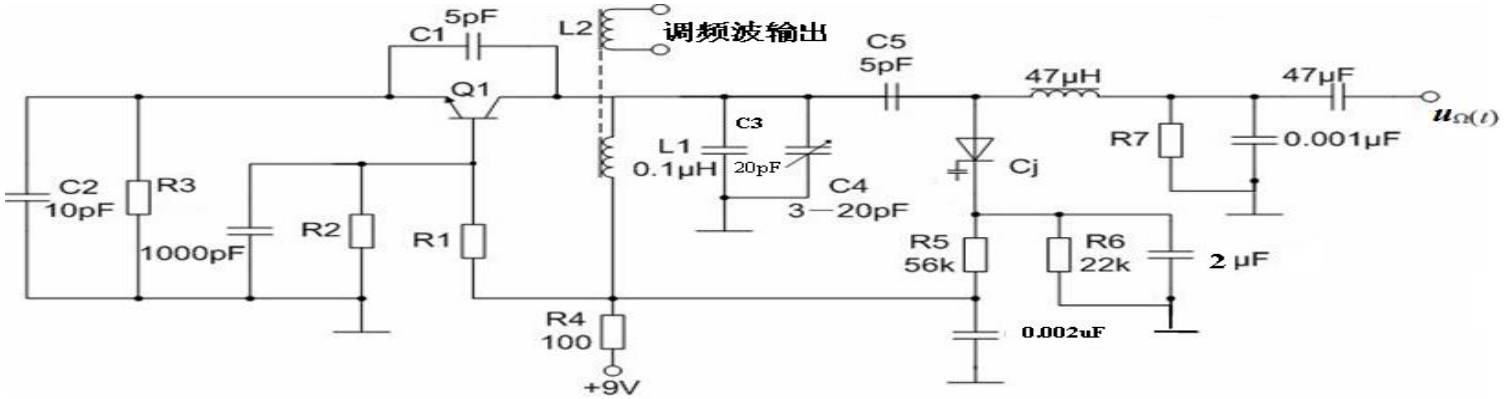
$R4$、R_5、R_6、R_7为变容管提供直流反偏置和直流通路;

1000pF，0.001μF、0.002μF均为高频旁路电容，即高频交流通路;

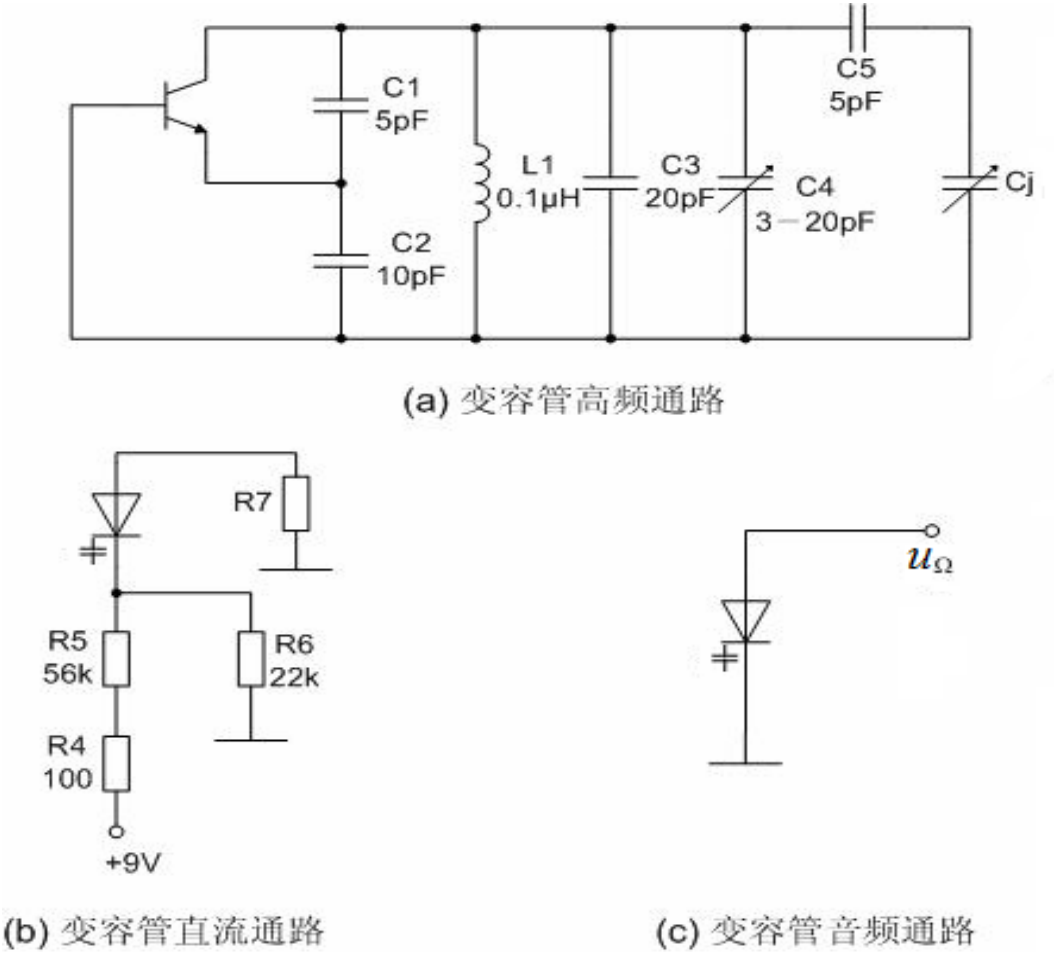
47μH为高频扼流图，对高频近似开路，对直流、低频近似短路;

47 μF、2 μF隔直电容，对低频近似短路;

C_1、C_2、C_3、C_4、C_5、C_j和L_1与晶体管Q1构成电容三点式振荡器； $L1$、$L2$互感耦合输出u_{FM}。



(2)



(a) 变容管高频通路

(b) 变容管直流通路

(c) 变容管音频通路

(3) $$U_Q \approx \frac{R_6}{R_5+R_6}\times 9\text{V} = \frac{22}{56+22}\times 9 = 2.54\text{V}$$

$$\therefore C_{jQ} = 80\Big/\sqrt{U_Q} = \frac{80}{\sqrt{2.54}} \approx 50\text{pF}$$

$$C_\Sigma = (C_1 串 C_2) 并 C_3 \ 并 C_4 \ 并 \ (C_5 串 C_j)$$

$$= \frac{C_1C_2}{C_1+C_2} + C_3 + C_4 + \frac{C_5C_j}{C_5+C_j}$$

$$\because \ f_c = \frac{1}{2\pi\sqrt{L_1C_\Sigma}} \qquad \therefore C_\Sigma = 31.3\times 10^{-12}F = 31.3\text{pF}$$

$$\therefore C_3 + C_4 = 23.5\text{pF}$$

$$P_1 = \frac{C_{\text{jQ}}}{C_5} = 10 \qquad P_2 = \frac{C_1 C_2 / (C_1 + C_2) + C_3 + C_4}{C_{\text{jQ}}} \approx 0.5$$

$$P = (1 + P_1)(1 + P_2 + P_1 P_2) = 71.5$$

由 $\Delta f_{\text{m}} = \dfrac{\gamma}{2P} \text{M} f_{\text{c}}$ 而 $\Delta f_m = m_{\text{f}} F = 5 \times 10^4 \text{Hz} = 50\text{kHz}$

$$50 \times 10^3 = \frac{\frac{1}{2}}{2 \times 71.5} \times \text{M} \times 90 \times 10^6$$

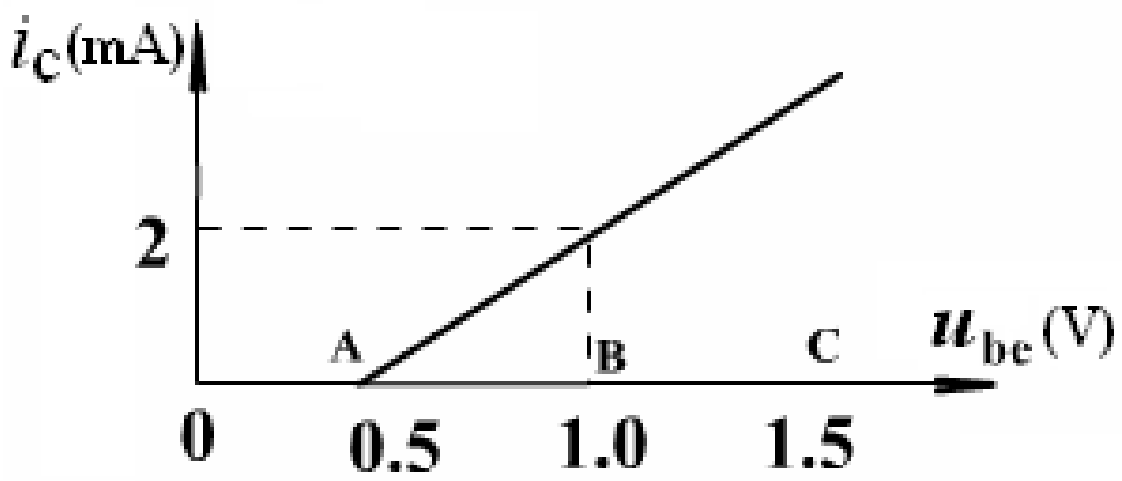
得 $\text{M} = 0.16$

又有$\mathrm{M}=\dfrac{U_{\Omega m}}{U_{\mathrm{Q}}+U_{\mathrm{D}}}\approx\dfrac{U_{\Omega m}}{U_{\mathrm{Q}}}$

得$U_{\Omega\mathrm{m}}\approx MU_{\mathrm{Q}}=0.16\times2.54=0.41\mathrm{V}$

混 频 器

例题：混频管的伏安特性如图所示，已知 $u_s(t) = 20\cos\omega_c t(\mathrm{mV}), u_L(t) = \cos\omega_L t(\mathrm{V})$。试分别求出工作点0，A, B, C四点时的混频跨导g_c。



g_m，$g_m(t)$, g_{m1}, g_c 四个跨导概念之间的关系和计算方法

解

$$O点：\ g_c = \frac{g_{m1}}{2} = \frac{1}{\pi}\int_0^{\pi/3} 4\cos\omega_L t d\omega_L t = \frac{2}{\pi}\sqrt{3}(\mathrm{ms})$$

A点：

$$\mathrm{g_c} = \frac{g_{m1}}{2} = \frac{1}{\pi}\int_0^{\pi/2} 4\cos\omega_L t d\omega_L t = \frac{4}{\pi}(\mathrm{ms})$$

B点：

$$g_\mathrm{c} = \frac{g_{m1}}{2} = \frac{1}{\pi}\int_0^{2\pi/3} 4\cos\omega_L t d\omega_L t = \frac{2}{\pi}\sqrt{3}(\mathrm{ms})$$

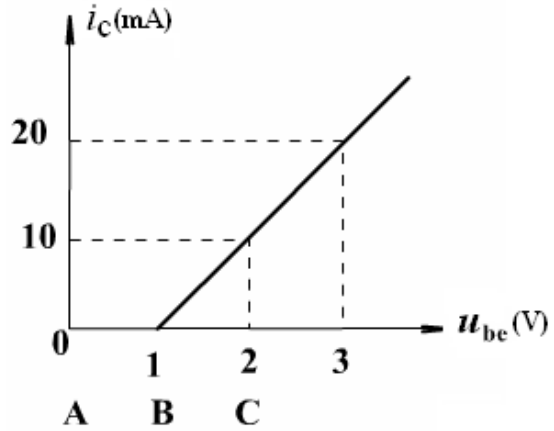
$$C点：\ g_c = 0$$

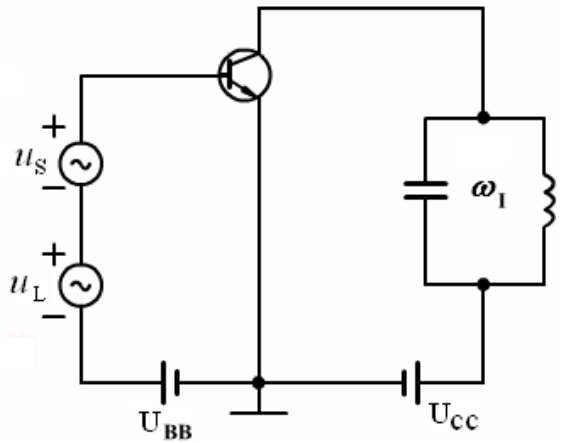
7-2.exe

例：在三极管混频器中管子的转移特性如图所示，

设中频$\omega_I = \omega_L - \omega_c$，$u_s(t) = 0.2\cos(\omega_c t + m_f \cos \Omega t)$(V)，

$u_L(t) = \cos \omega_L t$(V)，





试问 1.要使器件能正确工作，则工作点应取在 *A*、*B*、*C* 中哪一点?

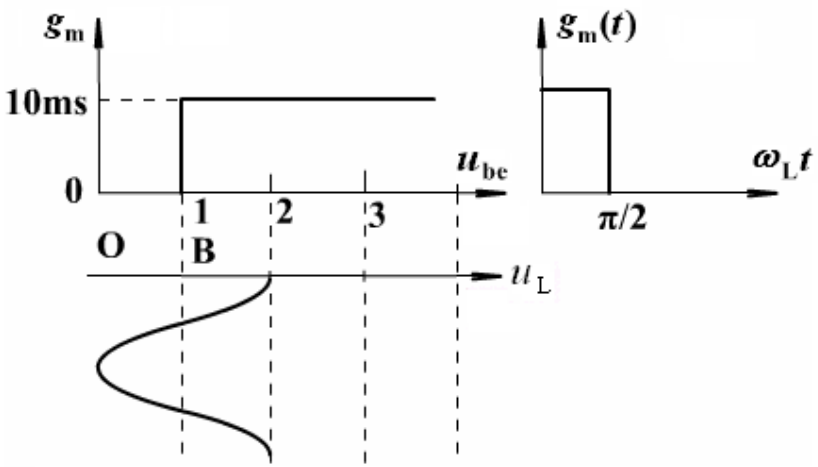
2.此时变频跨导为多少?

3. 写出中频电流表达式?

4. 若 $u_s(t) = 0.2(1 + 0.3\sin \Omega t)\cos \omega_c t$ (V) **时，写出中频电流表达式**

解

1.选B点



2. $$g_c = \frac{1}{\pi}\int_0^{\pi/2} 10\cos\omega_L t\, d\omega_L t$$

$$= \frac{1}{\pi}\times 10\text{ms}$$

$$= 3.2\text{ms}$$

3. $$i = U_{sm} g_c \cos(\omega_I t + m_f \cos\Omega t)$$

$$= 0.64(\cos\omega_I t + m_f \cos\Omega t) \qquad (\text{mA})$$

4. $$i = 0.64(1 + 0.3\sin\Omega t)\cos\omega_I t \qquad (\text{mA})$$

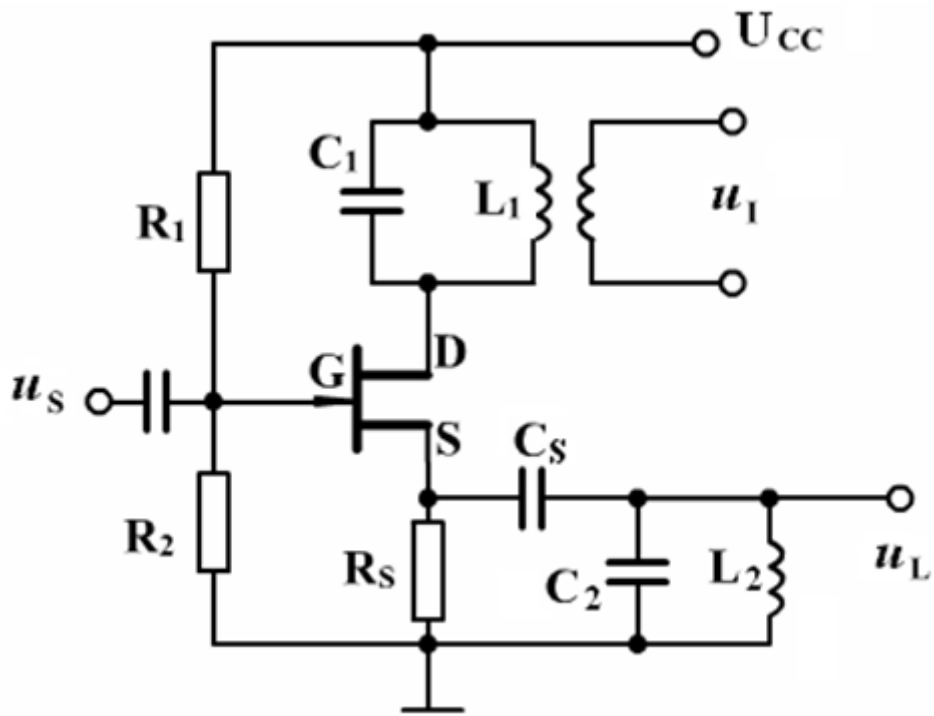
场效应管混频器

场效应管若工作在恒定区，其转移特性为平方律关系。例如，结型场效应管的漏级电流 i_D 表示为：

$$i_D = I_{DSS}\left(1-\frac{u_{GS}}{U_{GSoff}}\right)^2$$

I_{DSS}为栅源电压为零时的漏极电流，U_{GSoff}为夹断电压，u_{GS}为栅源电压。用场效应管作混频器时，由于它的平方律特性，不会产生高于二阶的谐波（短沟道效应），因而它的非线性失真比晶体管混频的小。

介绍结型场效应管混频器：



$$u_L = U_{Lm}\cos\omega_L t$$

$$u_S = U_{Sm}\cos\omega_c t$$

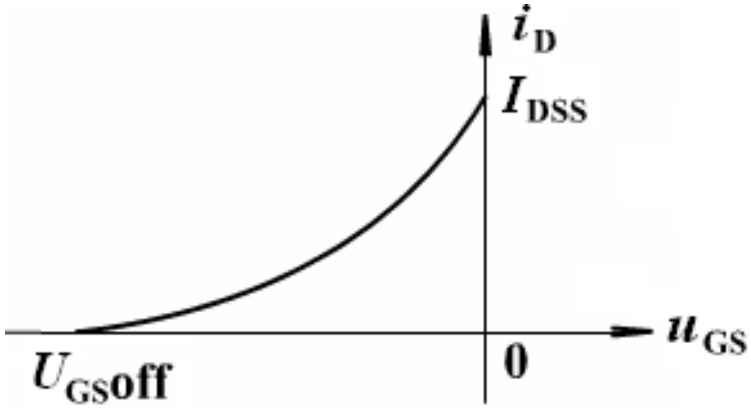
$$u_{GS} = U_{GS} + u_L + u_S$$

$$= U_{GS} + U_{Lm}\cos\omega_L t$$

$$+ \ U_{Sm}\cos\omega_c t$$

代入i_D的表达式之中

$$i_D = I_{DSS}(1-\frac{u_{GS}}{U_{GSoff}})^2$$



时变跨导$g_m(t)$为

$$g_{\mathrm{m}}(t)=\frac{d}{du_{\mathrm{GS}}}[I_{\mathrm{DSS}}(1-\frac{u_{\mathrm{GS}}}{U_{\mathrm{GSoff}}})^2]\Big|_{u_{\mathrm{GS}}=U_{\mathrm{GS}}+u_{\mathrm{L}}}$$

$$=-2I_{\mathrm{DSS}}(1-\frac{u_{GS}}{U_{GSoff}})\cdot(-\frac{1}{U_{GSoff}})\Big|_{u_{GS}=U_{GS}+u_L}$$

$$=-\frac{2I_{DSS}}{U_{GSoff}}(1-\frac{U_{GS}}{U_{GSoff}})+\frac{2I_{DSS}}{U_{GSoff}^2}U_{Lm}\cos\omega_L t$$

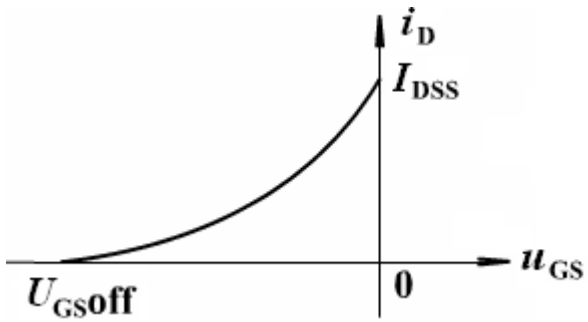
$$=g_{m0}+g_{m1}\cos\omega_L t$$

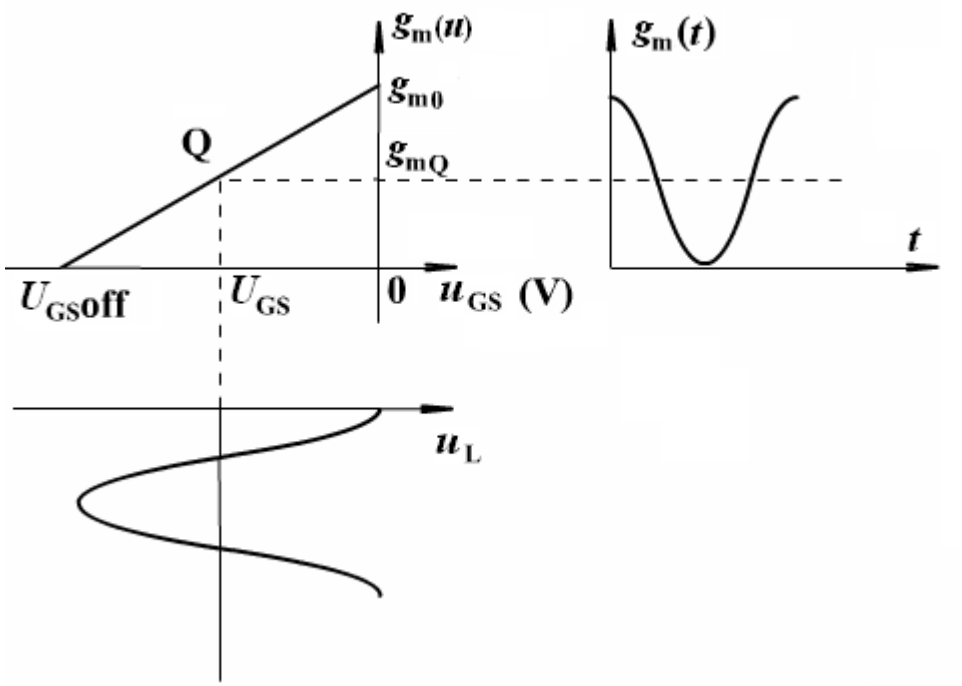
其混频跨导为：

$$g_{\mathrm{c}}=\frac{1}{2}g_{\mathrm{m1}}=\frac{I_{\mathrm{DSS}}}{U_{\mathrm{GSoff}}^2}U_{\mathrm{Lm}}$$

再用图解法来看如何求场效应管的时变跨导为：

$$g_m(u)=\frac{di_D}{du_{GS}}=-\frac{2I_{DSS}}{U_{GSoff}}(1-\frac{u_{GS}}{U_{GSoff}})$$





时变工作点：

$$u_{GS}=U_{GS}+u_L$$

$$U_{GS}=U_{LM}=\frac{U_{GSoff}}{2};$$

$$u_L=U_{LM}\cos\omega_L t$$

$$g_{m1} = g_{mQ} = \left.\frac{di_D}{du_{GS}}\right|_{u=U_{GS}} = \frac{I_{DSS}}{|U_{GSoff}|}$$

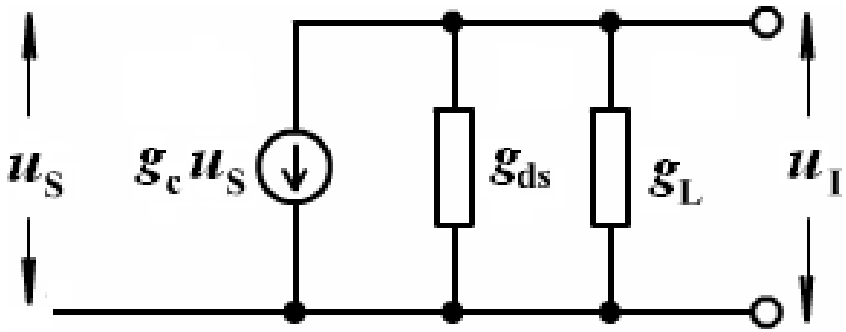
$$g_{c\max} = \frac{1}{2} g_{m1} = \frac{I_{DSS}}{2|U_{GSoff}|}$$

$$A_{uc} = \frac{u_{\mathrm{I}}}{u_{\mathrm{S}}} = \frac{g_{\mathrm{c}}}{g_{\mathrm{L}} + g_{\mathrm{ds}}}$$

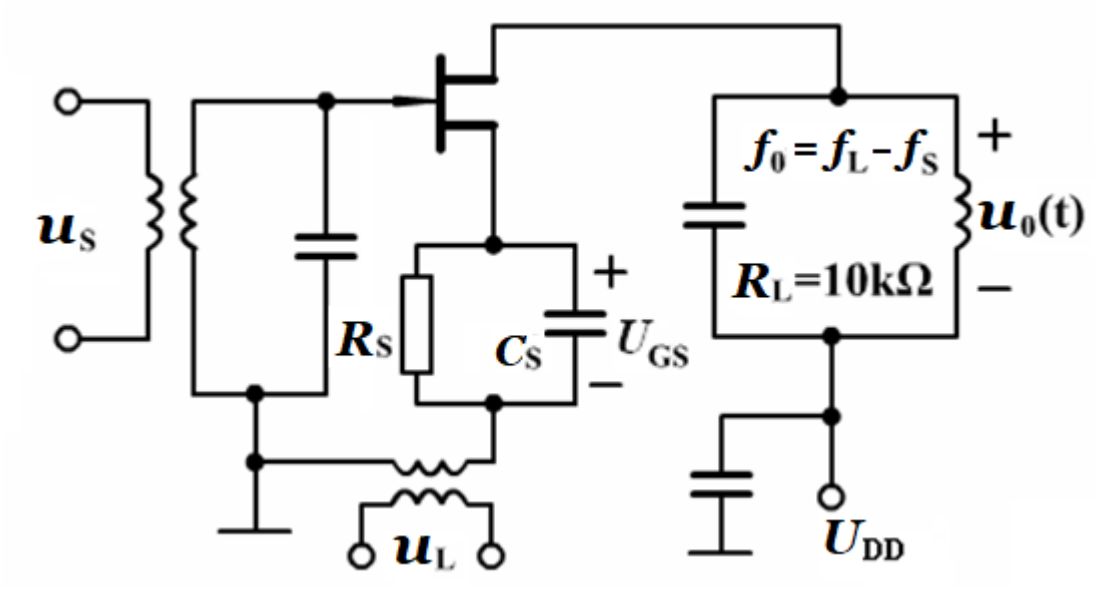
$$\approx g_{\mathrm{c}} R_{\mathrm{L}}$$

$$(g_{\mathrm{ds}} << g_{\mathrm{L}})$$

等效电路：

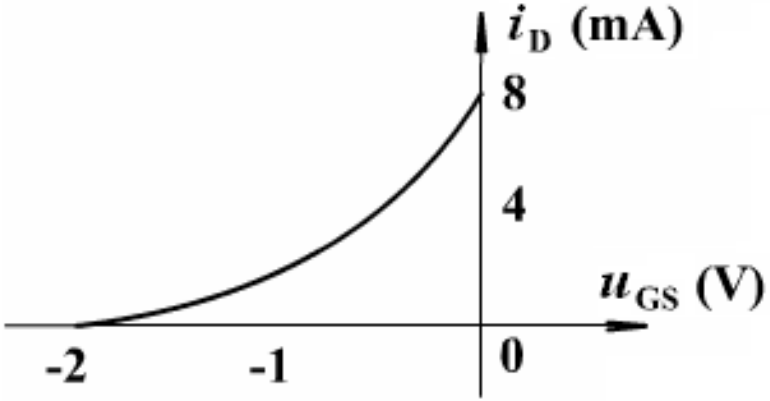


例：



场效应管转移特性为

$$i_D = I_{DSS}(1-\frac{u_{GS}}{U_{GSoff}})^2$$



1. 当$u_L=\cos\omega_L t$ (V) ,U_{GS}= -2(V)时画出$g_m(u)$ 及时变跨导$g_m(t)$ 的波形，并求出混频跨导g_c.

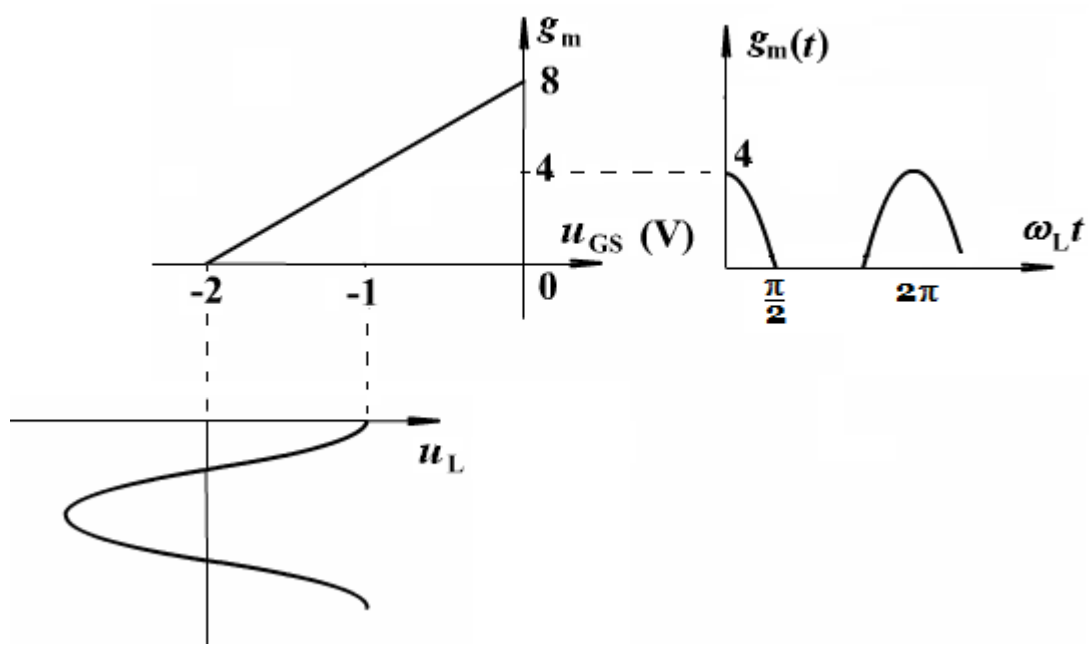
2. U_{GS}为多少时混频跨导最大g_c.

3. 条件同2，若输入电压$u_s=10\cos(\omega_c t+\Omega t)$(mV),求$u_o(t)$.

解 (1)

$$g_m(u) = \frac{di_D}{du_{GS}}$$

$$g_m(t) = \left.\frac{di_D}{du_{GS}}\right|_{u_{GS}=U_{GS}+u_L}$$

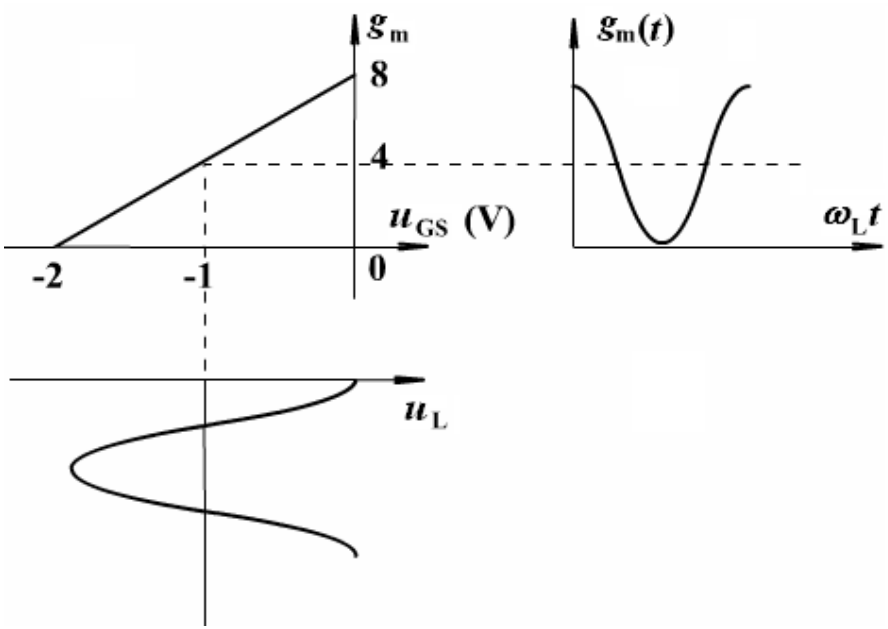


$$g_c = \frac{1}{2} g_{m1} = \frac{1}{\pi}\int_0^{\pi/2} 4\cos^2 \omega_L t d\omega_L t = 1\text{ms}$$

(2) U_{GS}= -1(V)

$$g_m(t) = g_{m0} + g_{m1}\cos\omega_L t$$

$$\therefore g_{c\max} = \frac{1}{2}g_{m1} = \frac{I_{DSS}}{2|U_{GSoff}|} = 2\text{ms}$$



(3)由于 g_c=2ms,R_L=10kΩ, 则$u_o(t)$为:

$$u_o(t) = g_c R_L u_S$$
$$= 2\times10\times10\cos(\omega_I - \Omega)t$$
$$= 0.2\cos(\omega_I - \Omega)t \quad (\text{V})$$